

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\A165QFA10CQ001\\plant air receiver7%.pvdb

Date of Analysis : Sep 18, 2026 2:08pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 5500.000 mm ID
 Vessel Design Length, Tangent to Tangent 16600.00 mm
 Distance of Bottom Tangent above Grade 3800.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 3500.00 mm
 Internal Design Temperature 85 °C
 Internal Design Pressure 1.100 MPa
 External Design Temperature 85 °C
 Maximum Allowable Working Pressure 1.15 MPa
 Shop Test Pressure 1.491 MPa
 Field Test Pressure 1.491 MPa
 Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -34.0 °C
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	+N
Head	SA-516 70	...	...	K02700	+N
Skirt	SA-285 C	...	...	K02801	
Skirt	SA-516 70	...	...	K02700	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 70	...	...	K02700	+N
Re-Pad	SA-516 70	...	...	K02700	+N
Nozzle Flg	SA-105	...	...	K03504	+N
Basering	SA-285 C	...	...	K02801	
Base Bolting	SA-36	...	...	K02600	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-2	...	...	...	1.5000	N/A	No
Skirt-1	...	...	...	1.5000	N/A	No
Btm_head	1.100	0.00	1.150	3.0000	No	No
Shell-1	1.100	0.00	1.150	3.0000	N/A	No
Shell-2	1.100	0.00	1.150	3.0000	N/A	No
Shell-3	1.100	0.00	1.150	3.0000	N/A	No
Shell-4	1.100	0.00	1.150	3.0000	N/A	No
Shell-5	1.100	0.00	1.150	3.0000	N/A	No
Shell-6	1.100	0.00	1.150	3.0000	N/A	No
Top_head	1.100	0.00	1.150	3.0000	No	No

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
--------------	--------------	-------------------	----------------------	-----------------------	----------------------	----------------------	----------	----------



Skirt	-1500.0	2000.0	14.0	14.0	...	...	0.70	0.70
Skirt	0.0	1500.0	14.0	14.0	...	...	0.70	0.70
Ellipse	50.0	50.0	30.0	26.0	24.9	...	1.00	1.00
Cylinder	3050.0	3000.0	26.0	26.0	25.0	...	1.00	1.00
Cylinder	6050.0	3000.0	26.0	26.0	25.0	...	1.00	1.00
Cylinder	9050.0	3000.0	26.0	26.0	25.0	...	1.00	1.00
Cylinder	12050.0	3000.0	26.0	26.0	25.0	...	1.00	1.00
Cylinder	15050.0	3000.0	26.0	26.0	25.0	...	1.00	1.00
Cylinder	16550.0	1500.0	26.0	26.0	25.0	...	1.00	1.00
Ellipse	16600.0	50.0	30.0	26.0	24.9	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N3_3"	WRC-107/537	0.895	n/a	Passed
N1_10"	WRC-297	0.993	Shell	Passed
N2_10"	WRC-107/537	0.895	n/a	Passed
N4_6"	WRC-107/537	0.895	n/a	Passed
N8_2"	WRC-107/537	0.895	n/a	Passed

Wind/Earthquake Shear, Bending:

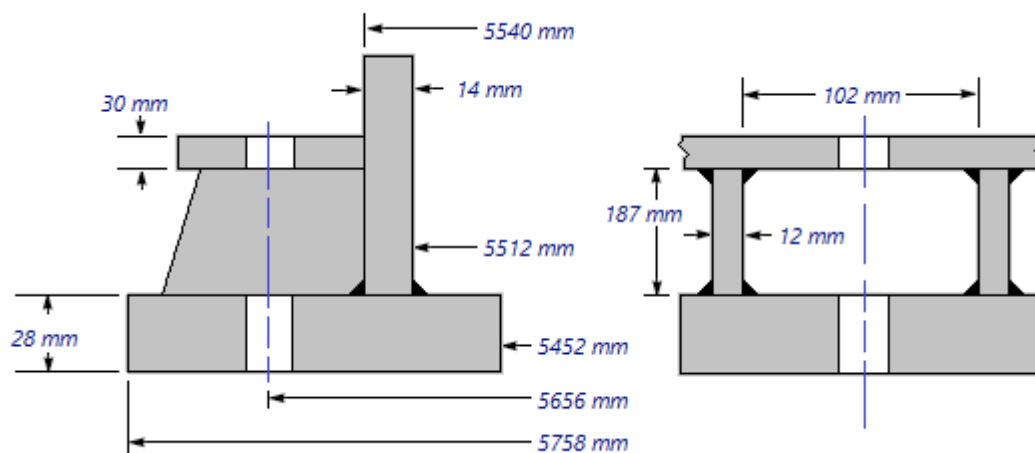
From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	244532	270434	3.144E+09	4.166E+09
20	30	2750	233701	269497	2.666E+09	3.626E+09
30	40	3525	225577	268004	2.321E+09	3.222E+09
40	50	5050	225306	262095	2.310E+09	3.209E+09
50	60	8050	195970	245543	1677503744	2.447E+09
60	70	11050	161313	224527	1141362816	1741868800
70	80	14050	122951	190213	714794688	1119507584
80	90	17050	103085	161885	375603264	591146688
90	100	19300	65004.6	103583	123366576	192783072
100	110	20075	54518.3	84097.2	33688028	51965520

Abs Max of the all of the Stress Ratio's : 0.4717

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	28.0000	mm
Inside Diameter of Basing	5452.0000	mm
Outside Diameter of Basing	5758.0000	mm
Nominal Diameter of Bolts	30.0000	mm
Diameter of Bolt Circle	5656.0000	mm
Number of Bolts	20	
Thickness of Gusset Plates	12.0000	mm
Average Width of Gusset Plates	109.0000	mm
Height of Gussets	187.0000	mm
Distance between Gussets	102.0000	mm
Thickness of Top Plate or Ring	30.0000	mm
Radial Width of the Top Plate	109.0000	mm
Number of Gussets	40	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	244532.	N
Total Earthquake Shear on Support	270434.	N
Wind Moment on Support	3222676736.	N-mm
Earthquake Moment on Support	4244424448.	N-mm
Maximum Initial Bolt Load	90513.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	95068.3	kgm
Shop Test - Fabricated + water (full)	532823.9	kgm
Shipping - Fab. + rem. intls.+ shipping app.	121992.9	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	121992.9	kgm
Empty - Fab. + intls. + details + wghts.	121992.9	kgm
Operating - Empty + operating liquid (no ca)	126894.9	kgm
Field Test - Empty weight + test medium (full)	559748.4	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N3_3"	1.15	...	...	OK	7.80	OK	Passed
N9B_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
N10B_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
M1B_24"	1.15	...	...	...		OK	Passed
N9A_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
N10A_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
M1A_24"	1.15	...	...	...		OK	Passed
N1_10"	1.15	...	...	OK	11.10	OK	Passed
N7_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
N5_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
N2_10"	1.15	...	...	OK	11.10	OK	Passed
N4_6"	1.15	...	...	OK	9.22	OK	Passed
N6_2"	1.15	...	...	OK	7.80	OK	No Calc[*]
N8_2"	1.15	...	...	OK	7.80	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1.15 Nozzle : N52"
 Minimum MAWP Shells/Flanges : 1.15 Element : Shell-6
 Minimum MAPnc Shells/Flanges : 1.3 Element : Shell-6

Computed Vessel M.A.W.P. : 1.15 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N3_3"	0.000	0.000	150.862
40	N9B_2"	250.000	180.000	123.724
40	N10B_2"	250.000	0.000	123.724
40	M1B_24"	1500.000	270.000	1127.200
60	N9A_2"	7500.000	180.000	123.724
60	N10A_2"	6999.999	0.000	123.724
60	M1A_24"	8000.000	270.000	1127.200
80	N1_10"	12999.999	180.000	505.293
80	N7_2"	12999.999	0.000	123.724
90	N5_2"	15049.999	0.000	123.724
100	N2_10"	0.000	0.000	505.293
100	N4_6"	0.000	180.000	310.145
100	N6_2"	0.000	0.000	123.724
100	N8_2"	0.000	45.000	123.724

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N9B_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N10B_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N9A_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N10A_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N7_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N5_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	226.27	150
N6_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	239.74	150
N8_2"	2.000 in	Actual	LWN	3.060	13.462	...	...	239.74	600
N3_3"	3.000 in	160	WNF	3.500	11.125	130.00	26.00	226.20	300
N4_6"	6.000 in	80	WNF	6.625	10.973	250.00	26.00	354.72	300
N1_10"	10.000 in	80	WNF	10.750	15.088	460.00	26.00	229.39	150
N2_10"	10.000 in	80	WNF	10.750	15.088	460.00	26.00	227.88	150
M1B_24"	24.000 in	Actual	WNF	24.000	26.000	1000.00	26.00	342.94	150
M1A_24"	24.000 in	Actual	WNF	24.000	26.000	1000.00	26.00	342.94	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N9B_2"	SA-106 B	26.000	10.000	...	...	...
N10B_2"	SA-106 B	26.000	10.000	...	...	...
N9A_2"	SA-106 B	26.000	10.000	...	...	...
N10A_2"	SA-106 B	26.000	10.000	...	...	...
N7_2"	SA-106 B	26.000	10.000	...	...	...
N5_2"	SA-106 B	26.000	10.000	...	...	...
N6_2"	SA-106 B	26.000	10.000	...	...	...
N8_2"	SA-106 B	26.000	10.000	...	...	...
N3_3"	SA-106 B	26.000	10.000	14.000	26.000	...
N4_6"	SA-106 B	26.000	10.000	20.000	26.000	...
N1_10"	SA-106 B	26.000	10.000	20.000	26.000	...
N2_10"	SA-106 B	26.000	10.000	20.000	26.000	...
M1B_24"	SA-516 70	26.000	10.000	20.000	26.000	...
M1A_24"	SA-516 70	26.000	10.000	20.000	26.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N9B_2"	250.000	180.0	200.00	0.00	Shell-1
N10B_2"	250.000	0.0	200.00	0.00	Shell-1
N9A_2"	7500.000	180.0	200.00	0.00	Shell-3
N10A_2"	6999.999	0.0	200.00	0.00	Shell-3
N7_2"	12999.999	0.0	200.00	0.00	Shell-5
N5_2"	15049.998	0.0	200.00	0.00	Shell-6
N6_2"	...	0.0	200.00	0.00	Top_head
N8_2"	...	45.0	200.00	0.00	Top_head
N3_3"	...	0.0	200.00	0.00	Btm_head



N4_6"	...	180.0	300.00	0.00	Top_head
N1_10"	12999.999	180.0	200.00	0.00	Shell-5
N2_10"	...	0.0	200.00	0.00	Top_head
M1B_24"	1500.000	270.0	300.00	0.00	Shell-1
M1A_24"	8000.000	270.0	300.00	0.00	Shell-3

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 30.0mm THK X 5500.0mm ID X 50.0mm	SA-516 70
4	PLATFORM: 1100mm X 1500mm WIDE	
1	LADDER: 4500.0mm	...
3	LADDER: 4000.0mm	...
1	PLATFORM: 1100mm X 7500mm WIDE	
1	LADDER: 6000.0mm	...
2	CLASS 150 GR 1.1, 24.0" BLIND FLANGE(S)	SA-105
1	BASERING: 5758.0mm OD X 5452.0 ID X 28.0mm	SA-285 C
20	BASERING BOLTS: 30.0mm DIA	SA-36
20	BASERING BOLT NUTS: 30.0mm DIA	...
1	BASERING TOP RING: 5758.0mm OD X 5540.0 ID X 30.0mm	...
40	BASERING GUSSET PLATES: 109.0mm X 109.0mm X 187.0mm X 12.0mm	...
5	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3032.5mm X 2000.0mm	SA-285 C
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2153.4mm X 2000.0mm	SA-285 C
5	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 3054.6mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SKIRT SEGMENT 14.0mm THK X 2043.2mm X 1500.0mm	SA-516 70
21	CYLINDRICAL SEGMENT 26.0mm THK X 3048.0mm X 3000.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 2038.7mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 26.0mm THK X 14230.5mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 26.0mm THK X 9143.9mm X 3000.0mm	SA-516 70
4	CYLINDRICAL SEGMENT 26.0mm THK X 3048.0mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 14230.5mm X 1500.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 26.0mm THK X 9143.9mm X 1500.0mm	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Btmhead	[10]	D	-34	-34		0.994	26.000	1.00	No
Btmhead	[7]	D	-30	-39		0.852	30.000	1.00	No
Shell-1	[8]	D	-34	-34		1.000	26.000	1.00	No
Shell-2	[8]	D	-34	-34		1.000	26.000	1.00	No
Shell-3	[8]	D	-34	-34		1.000	26.000	1.00	No
Shell-4	[8]	D	-34	-34		1.000	26.000	1.00	No
Shell-5	[8]	D	-34	-34		1.000	26.000	1.00	No
Shell-6	[8]	D	-34	-34		1.000	26.000	1.00	No
Tophead	[10]	D	-34	-34		0.994	26.000	1.00	No
Tophead	[7]	D	-30	-39		0.852	30.000	1.00	No
N33"	[1]	B	-34	-40		0.895	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N9B2"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N10B2"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
M1B24"	[1]	D	-34	-34		1.000	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N9A2"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N10A2"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
M1A24"	[1]	D	-34	-34		1.000	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N110"	[1]	B	-34	-34		1.000	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N72"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N52"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N210"	[1]	B	-34	-40		0.895	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N46"	[1]	B	-34	-40		0.895	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N62"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-41					
N82"	[1]	B	-19	-104	-29	0.027	13.462	1.00	No
Nozzle Flg	[5]	A	-18	-104					
Nozl. Bolting[22]			-48			1.000	64.000	1.00	No

Warmest MDMT: -18 -34

Required Minimum Design Metal Temperature -10.0 °C
 Warmest Computed Minimum Design Metal Temperature -34.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)



[20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 20 40 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	1.1	MPa
Design Internal Temperature	85.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-10.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW

Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm
Distance Upwind of Crest	Lh 0 mm
Distance from Crest to the Vessel	x 0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

M.A.W.P + Static Head to Nozzle	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No

Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



Element#	1/10
Element From Node	10
Element To Node	20
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	2000 mm
Skirt Outside Diameter	5540 mm
Diameter of Skirt at Base	5540 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-285 C
Allowable Stress, Ambient	108 MPa
Allowable Stress, Operating	108 MPa
Allowable Stress, Hydrotest	184.5 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	191.57 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	K02801
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	2/10
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1500 mm
Skirt Outside Diameter	5540 mm
Diameter of Skirt at Base	5540 mm
Skirt Thickness	14 mm
Internal Corrosion Allowance	1.5 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	85 °C
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 MPa
Allowable Stress, Operating	138 MPa
Allowable Stress, Hydrotest	234 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	242 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No

Element#	3/10
Element From Node	30
Element To Node	40
Element Type	Elliptical
Description	Btmhead
Distance "FROM" to "TO"	50 mm

Inside Diameter	5500	mm
Element Thickness	26	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	30	
Detail Type	Nozzle	
Detail ID	N33"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	3	in.
Nozzle Schedule	160	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	135.08	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	4/10	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	5500	mm
Element Thickness	26	mm
Internal Corrosion Allowance	3	mm
Nominal Thickness	26	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.1	MPa
Design Temperature Internal Pressure	85	°C
Design External Pressure	0	MPa
Design Temperature External Pressure	85	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	+N
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	500	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	8794560	mm ²
Platform Weight	58968	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4500	mm
Unit Weight of Ladder	0.49	N/mm



Platform Length (top head platform)	0	mm
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N9B2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	77.776	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	N10B2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	77.776	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	40	
Detail Type	Nozzle	
Detail ID	M1B24"	
Dist. from "FROM" Node / Offset dist	1450	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	5263	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +N	

Element#	5/10
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	1500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0

Platform Wind Area	8550720	mm ²
Platform Weight	30341	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1100	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	0	mm

Element#	6/10
Element From Node	60
Element To Node	70
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	2500 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	8550720 mm ²
Platform Weight	30341 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1100 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	2.0
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.49 N/mm
Platform Length (top head platform)	0 mm

Element From Node	60
Detail Type	Nozzle
Detail ID	N9A2"
Dist. from "FROM" Node / Offset dist	1450 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.776 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B



Element From Node	60
Detail Type	Nozzle
Detail ID	N10A2"
Dist. from "FROM" Node / Offset dist	950 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.776 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Nozzle
Detail ID	M1A24"
Dist. from "FROM" Node / Offset dist	1950 mm
Nozzle Diameter	24 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	5201.8 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70 +N

Element From Node	60
Detail Type	Weight
Detail ID	Clipwt
Dist. from "FROM" Node / Offset dist	2500 mm
Miscellaneous Weight	21279 N
Offset from Element Centerline	0 mm

Element#	7/10
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Shell-4
Distance "FROM" to "TO"	3000 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element#	8/10
Element From Node	80
Element To Node	90
Element Type	Cylinder
Description	Shell-5
Distance "FROM" to "TO"	3000 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa



Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	80
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	8550720 mm ²
Platform Weight	30341 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1100 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	2.0
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.49 N/mm
Platform Length (top head platform)	0 mm

Element From Node	80
Detail Type	Nozzle
Detail ID	N110"
Dist. from "FROM" Node / Offset dist	950 mm
Nozzle Diameter	10 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	656.33 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	80
Detail Type	Nozzle
Detail ID	N72"
Dist. from "FROM" Node / Offset dist	950 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.776 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	80
Detail Type	Weight
Detail ID	N1pipewt
Dist. from "FROM" Node / Offset dist	950 mm
Miscellaneous Weight	15513 N
Offset from Element Centerline	3903 mm

Element From Node	80
Detail Type	Weight
Detail ID	Trunnionwt
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	27869 N
Offset from Element Centerline	0 mm

Element#	9/10
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Shell-6
Distance "FROM" to "TO"	1500 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Nozzle
Detail ID	N52"
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	77.776 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element#	10/10
Element From Node	100
Element To Node	110
Element Type	Elliptical
Description	Tophead
Distance "FROM" to "TO"	50 mm
Inside Diameter	5500 mm
Element Thickness	26 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	30 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.1 MPa
Design Temperature Internal Pressure	85 °C
Design External Pressure	0 MPa
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70 +N
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Platform
Detail ID	PLAT:[1 OF 1]
Dist. from "FROM" Node / Offset dist	1500 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	0.0
Platform Wind Area	19426100 mm ²
Platform Weight	96758 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	7500 mm



Platform Height	1100	mm
Platform Clearance or End Offset	3750	mm
Platform Force Coefficient	2.0	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	6000	mm
Unit Weight of Ladder	0.49	N/mm
Platform Length (top head platform)	7500	mm

Element From Node	100	
Detail Type	Nozzle	
Detail ID	N210"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	10	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	656.33	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	100	
Detail Type	Nozzle	
Detail ID	N46"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	6	in.
Nozzle Schedule	80	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	383.57	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	100	
Detail Type	Nozzle	
Detail ID	N62"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	74.156	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	100	
Detail Type	Nozzle	
Detail ID	N82"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	600	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	88.639	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	100	
Detail Type	Weight	
Detail ID	N2Pipewt	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	12944	N
Offset from Element Centerline	3903	mm

Element From Node	100	
Detail Type	Weight	



Detail ID	PSVwt
Dist. from "FROM" Node / Offset dist	1350 mm
Miscellaneous Weight	19612 N
Offset from Element Centerline	3500 mm

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Skirt-2	...	...	2000	...	2000
Skirt-1	...	...	3500	...	1500
Btm_head	...	...	3550	...	50
Shell-1	...	...	6550	...	3000
Shell-2	...	...	9550	...	3000
Shell-3	...	...	12550	...	3000
Shell-4	...	...	15550	...	3000
Shell-5	...	...	18550	...	3000
Shell-6	...	...	20050	...	1500
Top_head	...	...	20100	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
Skirt-2		...	14	1.5	5540	...	1
Skirt-1		...	14	1.5	5540	...	2
Btm_head		1.1	30	3	5500	138	3
Shell-1		1.1	26	3	5500	138	4
Shell-2		1.1	26	3	5500	138	5
Shell-3		1.1	26	3	5500	138	6
Shell-4		1.1	26	3	5500	138	7
Shell-5		1.1	26	3	5500	138	8
Shell-6		1.1	26	3	5500	138	9
Top_head		1.1	30	3	5500	138	10

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-2		...	No Calc	No Calc	14	No Calc	1
Skirt-1		...	No Calc	No Calc	14	No Calc	2
Btm_head		1.1	1.15	1.3	26	24.9299	3
Shell-1		1.1	1.15	1.3	26	25.0497	4
Shell-2		1.1	1.15	1.3	26	25.0497	5
Shell-3		1.1	1.15	1.3	26	25.0497	6
Shell-4		1.1	1.15	1.3	26	25.0497	7
Shell-5		1.1	1.15	1.3	26	25.0497	8
Shell-6		1.1	1.15	1.3	26	25.0497	9
Top_head		1.1	1.15	1.3	26	24.9299	10

Minimum 1.15 1.3

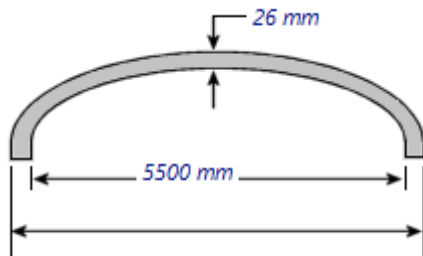
MAWP: 1.15 MPa, limited by: Shell-6.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btmhead

Elliptical Head Design Information:			
Design Pressure P	1.100 MPa	Int. Design Temperature	85 °C
Material	SA-516 70	Ext. Design Temperature	85 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	5500.000 mm	Actual Stress	131.6 MPa
Spec. Min. Thk t _e , or sa	26.0000 mm	Spec. Nominal Thk.	30.000 mm
Surface Area	34383248.0 mm ²	SG of Contents	0.000
Element Volume	22970.5 ltr	Weight of Contents	0.0 N
Empty Weight	85427.6 N	Operating Weight	85427.6 N
		Comp. All. Ope. Stress	103.6 MPa
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.375 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.1 \cdot 5506 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$

$$= 21.9299 + 3.0000 = 24.9299 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 23) / (1 \cdot 5506 + 0.2 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 26) / (1 \cdot 5500 + 0.2 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.1 \cdot (1 \cdot 5506 + 0.2 \cdot 23)) / (2 \cdot 1 \cdot 23)$$

$$= 131.584 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1) + 3$
 $= 25.050 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 27) / (2753 + 0.6 \cdot 27)$
 $= 1.35 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (5506 / (2 \cdot 1378))^2) / 6$
 $= 0.998549$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.368 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 26, tr = 22.9, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.99$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 30, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

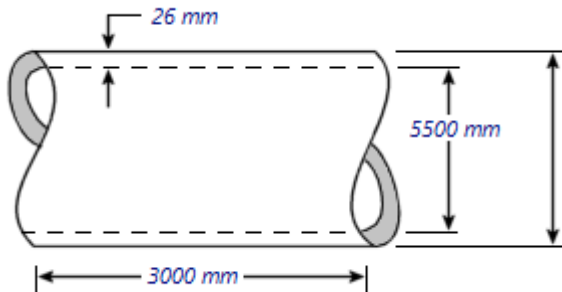
Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.375 mm

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		52326364.0 mm ²	SG of Contents		0.000
Element Volume		71287.6 ltr	Weight of Contents		0.0 N
Empty Weight		102909.1 N	Operating Weight		102909.1 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.471 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

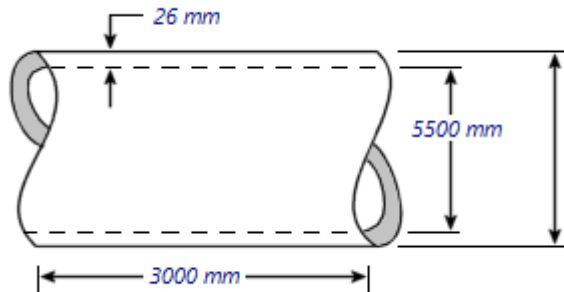
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t,e, or sa		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		52326364.0 mm²	SG of Contents		0.000
Element Volume		71287.6 ltr	Weight of Contents		0.0 N
Empty Weight		102909.1 N	Operating Weight		102909.1 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.471 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

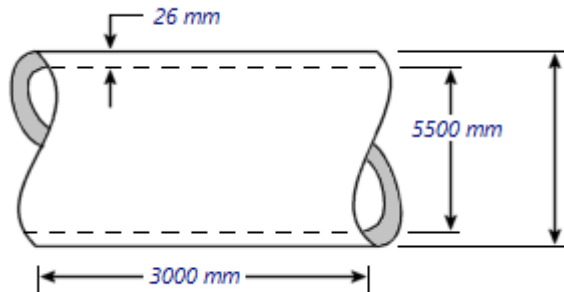
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		52326364.0 mm ²	SG of Contents		0.000
Element Volume		71287.6 ltr	Weight of Contents		0.0 N
Empty Weight		102909.1 N	Operating Weight		102909.1 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.471 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

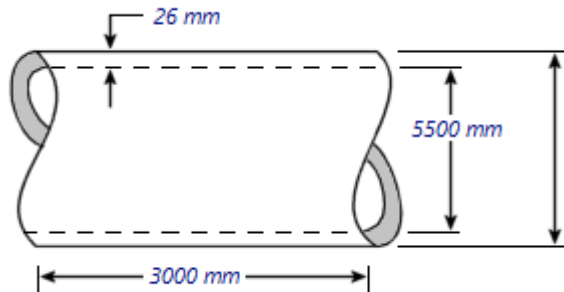
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		52326364.0 mm ²	SG of Contents		0.000
Element Volume		71287.6 ltr	Weight of Contents		0.0 N
Empty Weight		102909.1 N	Operating Weight		102909.1 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.471 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

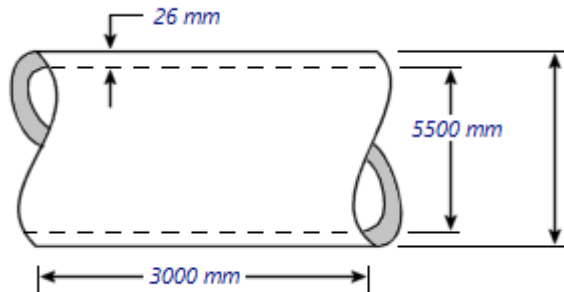
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		52326364.0 mm ²	SG of Contents		0.000
Element Volume		71287.6 ltr	Weight of Contents		0.0 N
Empty Weight		102909.1 N	Operating Weight		102909.1 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.471 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

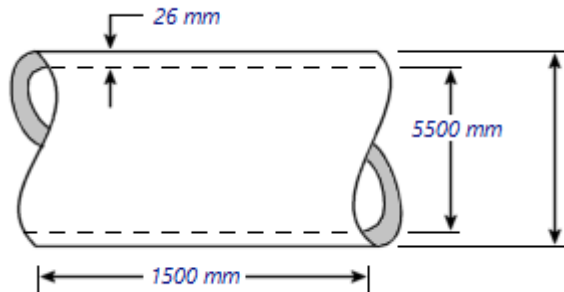
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	1.100 MPa	Int. Design Temperature		85 °C
Material		SA-516 70	Ext. Design Temperature		85 °C
Length		1500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 MPa
Inside Diameter	Di	5500.000 mm	Actual Stress		132.3 MPa
Spec. Min. Thk t,e, or sa		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		26163182.0 mm²	SG of Contents		0.000
Element Volume		35643.8 ltr	Weight of Contents		0.0 N
Empty Weight		51454.6 N	Operating Weight		51454.6 N
			Comp. All. Ope. Stress		103.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1)$$

$$= 22.0497 + 3.0000 = 25.0497 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 23) / (2753 + 0.6 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138 \cdot 1 \cdot 26) / (2750 + 0.6 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (1.1 \cdot (2753 + 0.6 \cdot 23)) / (1 \cdot 23)$$

$$= 132.325 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.471 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr * E* / (tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

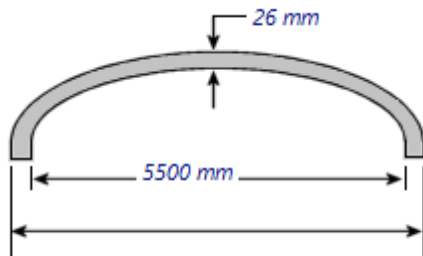
MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Material treatment options: +N

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Tophead

Elliptical Head Design Information:			
Design Pressure P	1.100 MPa	Int. Design Temperature	85 °C
Material	SA-516 70	Ext. Design Temperature	85 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.0000 mm	Allowable Stress (ope) S	138.0 MPa
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 MPa
Inside Diameter Di	5500.000 mm	Actual Stress	131.6 MPa
Spec. Min. Thk t _e , or sa	26.0000 mm	Spec. Nominal Thk.	30.000 mm
Surface Area	34383248.0 mm ²	SG of Contents	0.000
Element Volume	22970.5 ltr	Weight of Contents	0.0 N
Empty Weight	85427.6 N	Operating Weight	85427.6 N
		Comp. All. Ope. Stress	103.6 MPa
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.750 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.375 mm

Material treatment options: +N

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (1.1 \cdot 5506 \cdot 1) / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.1)$$

$$= 21.9299 + 3.0000 = 24.9299 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 23) / (1 \cdot 5506 + 0.2 \cdot 23)$$

$$= 1.15 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138 \cdot 1 \cdot 26) / (1 \cdot 5500 + 0.2 \cdot 26)$$

$$= 1.3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (1.1 \cdot (1 \cdot 5506 + 0.2 \cdot 23)) / (2 \cdot 1 \cdot 23)$$

$$= 131.584 \text{ MPa}$$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.1 \cdot 2753) / (138 \cdot 1 - 0.6 \cdot 1.1) + 3$
 $= 25.050 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (138 \cdot 1 \cdot 27) / (2753 + 0.6 \cdot 27)$
 $= 1.35 \text{ MPa}$

Factor K, corroded condition [Kcor]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (5506 / (2 \cdot 1378))^2) / 6$
 $= 0.998549$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.368 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 26, tr = 22.9, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.99$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 30, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.85$, Temp. Reduction = 8 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -30 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 °C

MDMT Design Information:			
MDMT Curve	D	Press. at MDMT	1.147 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	26.000 mm	Basic MDMT	-34 °C
Stress/Thk. Ratio	0.994	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 68.750 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 34.375 mm

Material treatment options: +N

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.491 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.430 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.633 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.262 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.573 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.573 MPa
App 27-4	= M.A.W.P.	1.147 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.15 • 1
 = 1.491 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	163.6	234.0	0.699	1.55
Shell-1	164.4	234.0	0.702	1.55
Shell-2	164.4	234.0	0.702	1.55
Shell-3	164.4	234.0	0.702	1.55
Shell-4	164.4	234.0	0.702	1.55
Shell-5	164.4	234.0	0.702	1.55
Shell-6	164.4	234.0	0.702	1.55
Top_head	163.6	234.0	0.699	1.55

Theoretical Maximum Allowable test pressure is : 2.146 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	118.00	118.00	1.000
N3_3"	Pad	138.00	138.00	1.000
N9B_2"	Nozzle	118.00	118.00	1.000
N10B_2"	Nozzle	118.00	118.00	1.000
M1B_24"	Nozzle	138.00	138.00	1.000
M1B_24"	Pad	138.00	138.00	1.000
N9A_2"	Nozzle	118.00	118.00	1.000
N10A_2"	Nozzle	118.00	118.00	1.000
M1A_24"	Nozzle	138.00	138.00	1.000
M1A_24"	Pad	138.00	138.00	1.000
N1_10"	Nozzle	118.00	118.00	1.000
N1_10"	Pad	138.00	138.00	1.000
N7_2"	Nozzle	118.00	118.00	1.000
N5_2"	Nozzle	118.00	118.00	1.000
N2_10"	Nozzle	118.00	118.00	1.000
N2_10"	Pad	138.00	138.00	1.000
N4_6"	Nozzle	118.00	118.00	1.000
N4_6"	Pad	138.00	138.00	1.000
N6_2"	Nozzle	118.00	118.00	1.000
N8_2"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000



Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_3"	6.44	216.00	0.030
N9B_2"	3.84	216.00	0.018
N10B_2"	3.84	216.00	0.018
M1B_24"	17.50	234.00	0.075
N9A_2"	3.84	216.00	0.018
N10A_2"	3.84	216.00	0.018
M1A_24"	17.50	234.00	0.075
N1_10"	15.36	216.00	0.071
N7_2"	3.84	216.00	0.018
N5_2"	3.84	216.00	0.018
N2_10"	15.36	216.00	0.071
N4_6"	12.92	216.00	0.060
N6_2"	3.84	216.00	0.018
N8_2"	3.84	216.00	0.018

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1.491 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1.430 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1.658 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	1.262 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	1.573 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1.573 MPa
App 27-4	= M.A.W.P.	1.147 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 1.15 • 1
 = 1.491 MPa

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm_head	201.1	234.0	0.859	1.68
Shell-1	200.5	234.0	0.857	1.67
Shell-2	197.0	234.0	0.842	1.64
Shell-3	193.5	234.0	0.827	1.61
Shell-4	189.9	234.0	0.812	1.58
Shell-5	186.4	234.0	0.797	1.55
Shell-6	182.8	234.0	0.781	1.52
Top_head	180.1	234.0	0.770	1.51

Theoretical Maximum Allowable test pressure is : 1.766 MPa

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N3_3"	Nozzle	118.00	118.00	1.000
N3_3"	Pad	138.00	138.00	1.000
N9B_2"	Nozzle	118.00	118.00	1.000
N10B_2"	Nozzle	118.00	118.00	1.000
M1B_24"	Nozzle	138.00	138.00	1.000
M1B_24"	Pad	138.00	138.00	1.000
N9A_2"	Nozzle	118.00	118.00	1.000
N10A_2"	Nozzle	118.00	118.00	1.000
M1A_24"	Nozzle	138.00	138.00	1.000
M1A_24"	Pad	138.00	138.00	1.000
N1_10"	Nozzle	118.00	118.00	1.000
N1_10"	Pad	138.00	138.00	1.000
N7_2"	Nozzle	118.00	118.00	1.000
N5_2"	Nozzle	118.00	118.00	1.000
N2_10"	Nozzle	118.00	118.00	1.000
N2_10"	Pad	138.00	138.00	1.000
N4_6"	Nozzle	118.00	118.00	1.000
N4_6"	Pad	138.00	138.00	1.000
N6_2"	Nozzle	118.00	118.00	1.000
N8_2"	Nozzle	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm_head	138.00	138.00	1.000



Shell-1	138.00	138.00	1.000
Shell-2	138.00	138.00	1.000
Shell-3	138.00	138.00	1.000
Shell-4	138.00	138.00	1.000
Shell-5	138.00	138.00	1.000
Shell-6	138.00	138.00	1.000
Top_head	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N3_3"	7.00	216.00	0.032
N9B_2"	4.15	216.00	0.019
N10B_2"	4.15	216.00	0.019
M1B_24"	18.88	234.00	0.081
N9A_2"	4.00	216.00	0.019
N10A_2"	4.00	216.00	0.019
M1A_24"	18.21	234.00	0.078
N1_10"	15.40	216.00	0.071
N7_2"	3.85	216.00	0.018
N5_2"	3.78	216.00	0.017
N2_10"	14.96	216.00	0.069
N4_6"	12.59	216.00	0.058
N6_2"	3.74	216.00	0.017
N8_2"	3.74	216.00	0.017

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	3500	...	...	No Calc	No Calc
20	30	3500	...	...	No Calc	No Calc
30	40	No Calc	5552	23	0.00057537	57.5239
40	50	17516.7	5552	23	0.00010728	10.7257
50	60	17516.7	5552	23	0.00010728	10.7257
60	70	17516.7	5552	23	0.00010728	10.7257
70	80	17516.7	5552	23	0.00010728	10.7257
80	90	17516.7	5552	23	0.00010728	10.7257
90	100	17516.7	5552	23	0.00010728	10.7257
100	110	No Calc	5552	23	0.00057537	57.5239

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	26	No Calc	...	0.26
40	50	26	No Calc	...	0.059
50	60	26	No Calc	...	0.059
60	70	26	No Calc	...	0.059
70	80	26	No Calc	...	0.059
80	90	26	No Calc	...	0.059
90	100	26	No Calc	...	0.059
100	110	26	No Calc	...	0.26

Minimum

0.059

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm^4	Ring Inertia Available mm^4
10	20	3500	No Calc	No Calc	No Calc
20	30	3500	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	17516.7	No Calc	No Calc	No Calc
50	60	17516.7	No Calc	No Calc	No Calc
60	70	17516.7	No Calc	No Calc	No Calc
70	80	17516.7	No Calc	No Calc	No Calc
80	90	17516.7	No Calc	No Calc	No Calc
90	100	17516.7	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	4877.71	...	4474.97	...	341.44
20	30	2825.58	...	2523.52	...	197.791
30	40	8711.77	22970.5	7840.59	23044.4	609.824
40	50	10494.5	71287.6	9288.63	71443.3	734.616
50	60	10494.5	71287.6	9288.63	71443.3	734.616
60	70	10494.5	71287.6	9288.63	71443.3	734.616
70	80	10494.5	71287.6	9288.63	71443.3	734.616
80	90	10494.5	71287.6	9288.63	71443.3	734.616
90	100	5247.25	35643.8	4644.32	35721.6	367.308
100	110	8711.77	22970.5	7840.59	23044.4	609.824
Total		82846	438022.97	73767	439026.84	5799

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
30	Nozl	14.7396	...	-1512	...	N3_3"
40	Plat	6434.36	0.32208E-04	500	...	PLAT:[1 OF 1]
40	Nozl	8.48673	-2850	200	...	N9B_2"
40	Nozl	8.48673	2850	200	...	N10B_2"
40	Nozl	574.286	...	1450	2900	M1B_24"
50	Plat	3310.75	...	1500	-2315.58	PLAT:[1 OF 1]
60	Plat	3310.75	...	2500	2315.58	PLAT:[1 OF 1]
60	Nozl	8.48673	-2850	1450	...	N9A_2"
60	Nozl	8.48673	2850	950	...	N10A_2"
60	Nozl	567.609	...	1950	2900	M1A_24"
60	Wght	2170	...	2500	...	Clip_wt
80	Plat	3310.75	...	500	-2315.58	PLAT:[1 OF 1]
80	Nozl	71.6164	-2850	950	...	N1_10"
80	Nozl	8.48673	2850	950	...	N7_2"
80	Wght	1582	-3903	950	...	N1_Pipe_wt
80	Wght	2842	...	1000	...	Trunnion_wt
90	Nozl	8.48673	2850	...	...	N5_2"
100	Plat	10557.9	...	1500	...	PLAT:[1 OF 1]
100	Nozl	71.6164	...	1512	...	N2_10"
100	Nozl	41.8538	-1500	1339.44	...	N4_6"
100	Nozl	8.09171	1500	1289.44	...	N6_2"
100	Nozl	9.67206	1060.66	1289.44	-1060.66	N8_2"
100	Wght	1320	3903	...	...	N2_Pipe_wt
100	Wght	2000	-3500	1350	...	PSV_wt

Total Weight of Each Detail Type:

Platforms	26924.6
Nozzles	1410.4
Weights	9914.0

Sum of the Detail Weights 38249.0 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	88645.9	88645.9	88645.9	88645.9	88645.9	88645.9
Nozzles	1410.4	1410.4	1410.4	1410.4	1410.4	1410.4
Wld Weights	5012.0	5012.0	5012.0	5012.0	5012.0	5012.0
Platforms	...	...	26924.6	26924.6	26924.6	26924.6
Ope Weights	...	...	...	...	...	4902.0
Test Liquid	...	437755.6	...	...	...	...

Totals	95068.3	532823.9	121992.9	121992.9	121992.9	126894.9
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Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	95068.3 kgm
Shop Test Wt.	- Fabricated weight + water (full)	532823.9 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	121992.9 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	121992.9 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	121992.9 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	126894.9 kgm
Field Test Wt.	- Empty weight + test medium (full)	559748.4 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	52253.0 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	224082.7 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	51143.3 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	34808848	34632920
20	30	26106636	25974690
30	40	34383248	33654492
40	50	52326364	51836276
50	60	52326364	51836276
60	70	52326364	51836276
70	80	52326364	51836276
80	90	52326364	51836276
90	100	26163182	25918138
100	110	34383248	33654492

Total	417476960.0	413016064.0 mm ²
Total	417.5	413.0 m ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	5219.15	5219.15	5219.15	...	5219.15
20	30	3023.37	3023.37	3023.37	...	3023.37
30	40	9336.33	9336.33	32292.8	...	9336.33
40	50	18254.7	18254.7	89498.9	2.02816	18254.7
50	60	14539.9	14539.9	85784	...	14539.9
60	70	17294.5	17294.5	88538.6	...	17294.5
70	80	11229.1	11229.1	82473.3	...	11229.1
80	90	17462	19044	88706.1	62337136	19044
90	100	5623.05	5623.05	41245.1	237275	5623.05
100	110	20010.8	23330.8	42967.2	18525396	23330.8

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
------	----	---	---------------------------------	----------------------------------



10	20	126895	126895	559748
20	30	121676	121676	554529
30	40	118652	118652	551506
40	50	109316	109316	519213
50	60	91061.3	91061.3	429714
60	70	76521.4	76521.4	343930
70	80	59226.9	59226.9	255392
80	90	47997.8	47997.8	172918
90	100	28953.8	28953.8	84212.3
100	110	23330.8	23330.8	42967.2

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	81099800	81099800	81099800
20	30	81099800	81099800	81099800
30	40	81099800	81099800	81099800
40	50	81099800	81099800	81099800
50	60	81099800	81099800	81099800
60	70	81099800	81099800	81099800
70	80	81099800	81099800	81099800
80	90	81099800	81099800	81099800
90	100	18762670	18762670	18762670
100	110	18525396	18525396	18525396

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Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
N3_3"	4.76	5.10	85	300	GR 1.1	4.877	2.268	4.763	2.268
N9B_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N10B_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
M1B_24"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N9A_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N10A_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
M1A_24"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N1_10"	1.81	1.96	85	150	GR 1.1	2.640	1.347	1.812	0.078
N7_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N5_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N2_10"	1.81	1.96	85	150	GR 1.1	2.640	1.347	1.812	0.079
N4_6"	4.76	5.10	85	300	GR 1.1	3.508	3.637	4.763	3.637
N6_2"	1.81	1.96	85	150	GR 1.1	...	...	...	...
N8_2"	9.53	10.20	85	600	GR 1.1	6.337	7.953	9.527	7.953
Min Rating	1.347	1.960, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N3_3"	111.125	0.50	1181217	4800	1.147	2.268	Pass
N1_10"	302.387	1.20	13125631	15999	1.147	1.347	Pass
N2_10"	302.387	1.20	13124623	15999	1.147	1.347	Pass
N4_6"	196.088	0.50	4724851	9600	1.147	3.637	Pass
N8_2"	77.851	0.50	524985	3200	1.147	7.953	Pass

Evaluating Flange on Nozzle [N33"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1181217) + 4 \cdot \text{abs}(4800)111.125 \leq \pi \cdot 111.125^3 \cdot ((4.763 - 1.147) + .500 \cdot 4.763)$$

$$= 0.21034E+08 \leq 0.25865E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N33" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 21033840.0 / (\pi \cdot 111.125^3))$$

$$= \min(4.763, 2.268)$$

$$= 2.268 \text{ MPa}$$

Evaluating Flange on Nozzle [N110"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(13125631) + 4 \cdot \text{abs}(15999)302.387 \leq \pi \cdot 302.387^3 \cdot ((1.812 - 1.147) + 1.200 \cdot 1.812)$$

$$= 0.22937E+09 \leq 0.24673E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N110" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1.812, 1.200 \cdot 1.812 + 1.812 - 229369792.0 / (\pi \cdot 302.387^3))$$

$$= \min(1.812, 1.347)$$

$$= 1.347 \text{ MPa}$$

Evaluating Flange on Nozzle [N210"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(13124623) + 4 \cdot \text{abs}(15999)302.387 \leq \pi \cdot 302.387^3 [(1.812 - 1.147) + 1.200 \cdot 1.812] \\ &= 0.22935E+09 \leq 0.24673E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N210" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.812, 1.200 \cdot 1.812 + 1.812 - 229353616.0 / (\pi \cdot 302.387^3)) \\ &= \min(1.812, 1.347) \\ &= 1.347 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N46"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(4724851) + 4 \cdot \text{abs}(9600)196.088 \leq \pi \cdot 196.088^3 [(4.763 - 1.147) + .500 \cdot 4.763] \\ &= 0.83130E+08 \leq 0.14211E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N46" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(4.763, 0.500 \cdot 4.763 + 4.763 - 83130120.0 / (\pi \cdot 196.088^3)) \\ &= \min(4.763, 3.637) \\ &= 3.637 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N82"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(524985) + 4 \cdot \text{abs}(3200)77.851 \leq \pi \cdot 77.851^3 [(9.527 - 1.147) + .500 \cdot 9.527] \\ &= 0.93966E+07 \leq 0.19488E+08 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N82" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(9.527, 0.500 \cdot 9.527 + 9.527 - 9396616.0 / (\pi \cdot 77.851^3)) \\ &= \min(9.527, 7.953) \\ &= 7.953 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 6.02944 Hz.

The Natural Frequency for the Vessel (Ope...) is 6.02944 Hz.

The Natural Frequency for the Vessel (Filled) is 3.04832 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.15 \cdot 0.92) / (1 + 1.7 \cdot 3.4 \cdot 0.15)))$
 $= \min(0.85, 0.89)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	6.029 Hz
Natural Frequency of Vessel (Empty)	6.029 Hz
Natural Frequency of Vessel (Test)	3.048 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.549
Structure Height to Diameter ratio	3.938
Height to top of Structure	21500.998 mm

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Zg = 589788$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 1000 + 300 = 1300$ mm
 $= 4.27$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (4.27 ft.) < 15 ft.
 $= 2.41(15 / Zg)^{2/\alpha}$
 $= 2.41(15 / 1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [1.59 \text{ kPa}]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.55 \cdot 143$$

$$= 2214 \text{ lbs. } [9847 \text{ N}]$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-2	1300.0	0.000	0.000	0.000	1.035	1.000	1.587
Skirt-1	3050.0	0.000	0.000	0.000	1.035	1.000	1.587
Btmhead	3825.0	0.000	0.000	0.000	1.035	1.000	1.587
Shell-1	5350.0	0.000	0.000	0.000	1.064	1.000	1.631
Shell-2	8350.0	0.000	0.000	0.000	1.149	1.000	1.762
Shell-3	11350.0	0.000	0.000	0.000	1.212	1.000	1.859
Shell-4	14350.0	0.000	0.000	0.000	1.263	1.000	1.936
Shell-5	17350.0	0.000	0.000	0.000	1.305	1.000	2.001
Shell-6	19600.0	0.000	0.000	0.000	1.333	1.000	2.044
Tophead	20967.7	0.000	0.000	0.000	1.349	1.000	2.068

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(10^4)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)
- Vp - Vib. Chance, <= 0.31416E-05 (High); 0.31416E-05 < 0.39270E-05 (Probable)
- P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5
 :: 0/20100 = 0

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)
 :: $[18.2/65.9^2] \cdot 10^4 = 41.9$ [Geometry Violation]

Compute the vibration possibility. If Vp > 0.39270E-05 no chance. [Vp]:

$$= W / (L \cdot Dr^2)$$

$$= 1149064 / (20100 \cdot 5552^2)$$

$$= 0.18546E-05$$

Compute the damping factor Df which is a measure of instability [Df]:



$$= W \cdot DI / (L \cdot Dr^2)$$

$$= 1149064 \cdot 0.03 / (20100 \cdot 5552^2)$$

$$= 0.55638E-07$$

Compute the critical wind velocity [Vc]:

$$= 3.4 \cdot f \cdot Dr$$

$$= 3.4 \cdot 6.03 \cdot 18.2$$

$$= 373.414 \text{ mile/hr } [600.935 \text{ km/hr}]$$

Compute the velocity at the top of the tower [Vw]:

$$= V30 \cdot (L / (30 + \text{BaseHeight}))^{0.143}$$

$$= 112 \cdot (65.9 / (30 + 0.98))^{0.143}$$

$$= 124.608 \text{ mile/hr } [200.531 \text{ km/hr}]$$

Compute the maximum gust velocity using the gust response factor Gh [Vg]:

$$= Vw \cdot Gh$$

$$= 201 \cdot 0.85$$

$$= 170.452 \text{ km/hr}$$

Since Vc is greater than Vg the dynamic deflection Z, does not need to be computed.

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
PLAT:[1 OF 1]	8794560.00	4350.00	1.60	14090.96	2.00
PLAT:[1 OF 1]	8550720.00	8350.00	1.76	15069.49	2.00
PLAT:[1 OF 1]	8550720.00	12350.00	1.88	16116.41	2.00
PLAT:[1 OF 1]	8550720.00	16350.00	1.98	16928.11	2.00
PLAT:[1 OF 1]	19426100.00	21850.00	2.08	40462.24	2.00

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
Clipwt	0.00	12350.00	1.88	0.00
N1pipewt	0.00	16800.00	1.99	0.00
Trunnionwt	0.00	16850.00	1.99	0.00
N2pipewt	0.00	20350.00	2.06	0.00
PSVwt	0.00	21700.00	2.08	0.00

The Natural Frequency for the Vessel (Ope...) is 6.02944 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	1300	6648	13296001	1.58714	10831.4
20	30	3050	6648	9972000	1.58714	8123.58
30	40	3825	6662.4	333120	1.58714	271.372
40	50	5350	6662.4	19987200	1.63111	29336.3
50	60	8350	6662.4	19987200	1.76241	34656.9
60	70	11350	6662.4	19987200	1.85905	38361.9
70	80	14350	6662.4	19987200	1.93644	19865.9
80	90	17350	6662.4	19987200	2.00144	38080.4
90	100	19600	6662.4	9993600	2.04434	10486.4
100	110	20967.7	6662.4	7664044	2.06846	54518.3

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.





Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	1.500
Site Class	D
Component Elevation Ratio	z/h
Amplification Factor	Ap
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR]
Component strength factor	[Rpo]
Overstrength factor	[Omega0]
Response Acceleration Factor	[SMS]
Response Acceleration Factor	[SM1]
Building Importance Factor	[Ie]
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 $= Ct \cdot h_{nx}$ where $Ct = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 $= 0.020(70.50.75)$
 $= 0.486$ seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 $= (1/\text{Natural Frequency}) = (1/6.03)$
 $= 0.166$

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 $= \text{Minimum Value of } (1.7 \cdot 0.49, 0.17)$ per 12.8.2
 $= 0.166$

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 $= Sds / (R/I)$
 $= 0.22 / (1.5/1.5)$
 $= 0.217$

Maximum value of C_s per equation 12.8-3 [Cs]:
 $= Sd1 / (T(R/Ie))$
 $= 0.093 / (0.17 (1.5/1.5))$
 $= 0.559$

Minimum value of C_s per equation 15.4-1 [Cs]:
 $= \max(0.044 \cdot SDS \cdot Ie, 0.030)$
 $= \max(0.044 \cdot 0.22 \cdot 1.5, 0.030)$
 $= 0.030$

Total Base Shear [V]:
 $= C_s \cdot W$ (Equation 12.8-1):
 $= 0.22 \cdot 1244330$
 $= 270434.500$ N

Vertical load per 12.4-4, [Ev]:
 $= 0.2 \cdot Sds \cdot W$
 $= 0.2 \cdot 0.22 \cdot 1244330$



= 54086.9 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x^k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1

The Natural Frequency for the Vessel (Ope...) is 6.02944 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	1000	51179	937.123	960.635	54086.9
20	30	2750	29647.1	1492.87	1530.32	51862.3
30	40	3525	91552	5909.25	6057.51	50573.7
40	50	5050	179006	16552.5	16967.8	46594.2
50	60	8050	142578	21016.2	21543.5	38813.4
60	70	11050	169589	34313.6	35174.5	32616
70	80	14050	110113	28328.2	29039	25244.5
80	90	17050	186745	58301.4	54799.5	20458.3
90	100	19300	55139.6	19486.2	19975.1	12341.1
100	110	20075	228781	84097.2	73939.8	9944.37

The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	1000	244532	270434	3.144E+09	4.166E+09
20	30	2750	233701	269497	2.666E+09	3.626E+09
30	40	3525	225577	268004	2.321E+09	3.222E+09
40	50	5050	225306	262095	2.310E+09	3.209E+09
50	60	8050	195970	245543	1677503744	2.447E+09
60	70	11050	161313	224527	1141362816	1741868800
70	80	14050	122951	190213	714794688	1119507584
80	90	17050	103085	161885	375603264	591146688
90	100	19300	65004.6	103583	123366576	192783072
100	110	20075	54518.3	84097.2	33688028	51965520

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	244532	0.008756	0.034107	0.33195E-04
20	30	233701	0.063196	0.1003	0.54569E-04
30	40	225577	0.10167	0.10304	0.54950E-04
40	50	225306	0.1936	0.29885	0.74552E-04
50	60	195970	0.41653	0.54457	0.88377E-04
60	70	161313	0.68107	0.82435	0.97441E-04
70	80	122951	0.97293	1.12551	0.00010277
80	90	103085	1.28087	1.438	0.00010514
90	100	65004.6	1.51696	1.59605	0.00010552
100	110	54518.3	1.59869	1.60133	0.00010552

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	719.563	4497.27
20	30	719.563	4497.27
30	40	721.122	4507.01
40	50	721.122	4507.01
50	60	721.122	4507.01
60	70	721.122	4507.01
70	80	721.122	4507.01
80	90	721.122	4507.01
90	100	721.122	4507.01
100	110	721.122	4507.01

Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
 Allowable deflection : 100.500 Actual deflection : 1.601 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	243045	217065	334921216	299281920
20	30	243045	217065	334921216	299281920
30	40	451369	399505	620662400	549939264
40	50	451369	399505	620662400	549939264
50	60	451369	399505	620662400	549939264
60	70	451369	399505	620662400	549939264
70	80	451369	399505	620662400	549939264
80	90	451369	399505	620662400	549939264
90	100	451369	399505	620662400	549939264
100	110	451369	399505	620662400	549939264

Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	90.7184	154.977	-67.6743	-67.6743
20	30	115.918	196.556	-67.6743	-67.6743
30	40	165.6	280.8	-103.644	-103.644
40	50	165.6	280.8	-103.644	-103.644
50	60	165.6	280.8	-103.644	-103.644
60	70	165.6	280.8	-103.644	-103.644
70	80	165.6	280.8	-103.644	-103.644
80	90	165.6	280.8	-103.644	-103.644
90	100	165.6	280.8	-103.644	-103.644
100	110	165.6	280.8	-103.644	-103.644

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Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	65.6126	...	88.9543
40	50	65.6126	...	88.9543
50	60	65.6126	...	88.9543
60	70	65.6126	...	88.9543
70	80	65.6126	...	88.9543
80	90	65.6126	...	88.9543
90	100	65.6126	...	88.9543
100	110	65.6126	...	88.9543

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-5.73302	-5.73302	-25.289	0.27089	0.27089
20	30	-5.49722	-5.49722	-25.0532	0.27089	0.27089
30	40	-2.91261	-2.91261	-2.91261	0.14742	0.14742
40	50	-2.68343	-2.68343	-2.68343	0.14742	0.14742
50	60	-2.23532	-2.23532	-2.23532	0.14742	0.14742
60	70	-1.8784	-1.8784	-1.8784	0.14742	0.14742
70	80	-1.45387	-1.45387	-1.45387	0.14742	0.14742
80	90	-1.17822	-1.17822	-1.17822	0.14742	0.14742
90	100	-0.71074	-0.71074	-0.71074	0.034107	0.034107
100	110	-0.57271	-0.57271	-0.57271	0.033675	0.033675

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.27089	10.5017	13.9146	3.46556	...
20	30	0.27089	8.90363	12.1104	2.9382	...
30	40	0.14742	4.21902	5.85747	1.39228	...
40	50	0.14742	4.19852	5.83337	1.38551	...
50	60	0.14742	3.04937	4.44863	1.00629	...
60	70	0.14742	2.07477	3.16637	0.68467	...
70	80	0.14742	1.29935	2.03504	0.42879	...
80	90	0.14742	0.68277	1.07459	0.22531	...
90	100	0.034107	0.22426	0.35044	0.074004	...
100	110	0.033675	0.061238	0.094463	0.020209	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	13.1336

20	30	...	...	...	11.3993
30	40	...	...	...	5.4991
40	50	...	...	...	5.47597
50	60	...	...	...	4.1505
60	70	...	...	...	2.93008
70	80	...	...	...	1.86438
80	90	...	...	...	0.97384
90	100	...	...	...	0.31199
100	110	...	...	...	0.083053

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	0.2492
20	30	...	0.23895
30	40	...	0.1266
40	50	...	0.11664
50	60	...	0.097162
60	70	...	0.081648
70	80	...	0.063195
80	90	...	0.051213
90	100	...	0.030894
100	110	...	0.024894

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.04	90.72	-16.51	67.67	0.0556	0.2439
20	3.68	115.92	-14.67	67.67	0.0317	0.2168
30	1.45	165.60	-7.28	103.64	0.0088	0.0702
40	1.66	165.60	-7.03	103.64	0.0100	0.0678
50	0.96	165.60	-5.43	103.64	0.0058	0.0524
60	0.34	165.60	-4.10	103.64	0.0021	0.0396
70		165.60	-2.90	103.64		0.0280
80		165.60	-2.01	103.64		0.0194
90		165.60	-0.97	103.64		0.0094
100		165.60	-0.67	103.64		0.0064

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	7.92	90.72	-18.89	67.67	0.0873	0.2791
20	6.41	115.92	-16.93	67.67	0.0553	0.2501
30	2.86	165.60	-8.43	103.64	0.0173	0.0814
40	3.06	165.60	-8.19	103.64	0.0185	0.0790
50	2.16	165.60	-6.44	103.64	0.0130	0.0621
60	1.28	165.60	-4.87	103.64	0.0077	0.0470
70	0.62	165.60	-3.40	103.64	0.0038	0.0328
80		165.60	-2.25	103.64		0.0217
90		165.60	-1.03	103.64		0.0099

100 165.60 -0.66 103.64 0.0064

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.04	90.72	-16.51	67.67	0.0556	0.2439
20	3.68	115.92	-14.67	67.67	0.0317	0.2168
30	1.45	165.60	-7.28	103.64	0.0088	0.0702
40	1.66	165.60	-7.03	103.64	0.0100	0.0678
50	0.96	165.60	-5.43	103.64	0.0058	0.0524
60	0.34	165.60	-4.10	103.64	0.0021	0.0396
70		165.60	-2.90	103.64		0.0280
80		165.60	-2.01	103.64		0.0194
90		165.60	-0.97	103.64		0.0094
100		165.60	-0.67	103.64		0.0064

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.70	90.72	-19.67	67.67	0.0959	0.2906
20	7.12	115.92	-17.64	67.67	0.0614	0.2607
30	3.22	165.60	-8.79	103.64	0.0194	0.0848
40	3.41	165.60	-8.55	103.64	0.0206	0.0825
50	2.46	165.60	-6.73	103.64	0.0148	0.0650
60	1.52	165.60	-5.11	103.64	0.0092	0.0493
70	0.79	165.60	-3.57	103.64	0.0048	0.0345
80	0.09	165.60	-2.35	103.64	0.0006	0.0227
90		165.60	-1.06	103.64		0.0103
100		165.60	-0.68	103.64		0.0065

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-29.03	67.67		0.4289
20		196.56	-28.26	67.67		0.4176
30		280.80	-4.45	103.64		0.0430
40		280.80	-4.22	103.64		0.0407
50		280.80	-3.39	103.64		0.0327
60		280.80	-2.71	103.64		0.0262
70		280.80	-2.03	103.64		0.0196
80		280.80	-1.55	103.64		0.0150
90		280.80	-0.82	103.64		0.0079
100		280.80	-0.63	103.64		0.0060

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-25.56	67.67		0.3777
20		196.56	-25.32	67.67		0.3742
30		280.80	-3.06	103.64		0.0295
40		280.80	-2.83	103.64		0.0273
50		280.80	-2.38	103.64		0.0230
60		280.80	-2.03	103.64		0.0195
70		280.80	-1.60	103.64		0.0154
80		280.80	-1.33	103.64		0.0128
90		280.80	-0.74	103.64		0.0072
100		280.80	-0.61	103.64		0.0059

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.04	90.72	-16.51	67.67	0.0556	0.2439
20	3.68	115.92	-14.67	67.67	0.0317	0.2168
30	67.07	165.60		103.64	0.4050	
40	67.28	165.60		103.64	0.4063	



50	66.57	165.60	103.64	0.4020
60	65.96	165.60	103.64	0.3983
70	65.61	165.60	103.64	0.3962
80	65.26	165.60	103.64	0.3941
90	65.16	165.60	103.64	0.3935
100	65.13	165.60	103.64	0.3933

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.70	90.72	-19.67	67.67	0.0959	0.2906
20	7.12	115.92	-17.64	67.67	0.0614	0.2607
30	68.83	165.60		103.64	0.4156	
40	69.03	165.60		103.64	0.4168	
50	68.07	165.60		103.64	0.4111	
60	67.13	165.60		103.64	0.4054	
70	66.40	165.60		103.64	0.4010	
80	65.71	165.60		103.64	0.3968	
90	65.32	165.60		103.64	0.3944	
100	65.19	165.60		103.64	0.3937	

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	5.04	90.72	-16.51	67.67	0.0556	0.2439
20	3.68	115.92	-14.67	67.67	0.0317	0.2168
30	1.45	165.60	-7.28	103.64	0.0088	0.0702
40	1.66	165.60	-7.03	103.64	0.0100	0.0678
50	0.96	165.60	-5.43	103.64	0.0058	0.0524
60	0.34	165.60	-4.10	103.64	0.0021	0.0396
70		165.60	-2.90	103.64		0.0280
80		165.60	-2.01	103.64		0.0194
90		165.60	-0.97	103.64		0.0094
100		165.60	-0.67	103.64		0.0064

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	8.70	90.72	-19.67	67.67	0.0959	0.2906
20	7.12	115.92	-17.64	67.67	0.0614	0.2607
30	3.22	165.60	-8.79	103.64	0.0194	0.0848
40	3.41	165.60	-8.55	103.64	0.0206	0.0825
50	2.46	165.60	-6.73	103.64	0.0148	0.0650
60	1.52	165.60	-5.11	103.64	0.0092	0.0493
70	0.79	165.60	-3.57	103.64	0.0048	0.0345
80	0.09	165.60	-2.35	103.64	0.0006	0.0227
90		165.60	-1.06	103.64		0.0103
100		165.60	-0.68	103.64		0.0065

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		154.98	-29.03	67.67		0.4289
20		196.56	-28.26	67.67		0.4176
30	87.58	280.80		103.64	0.3119	
40	87.80	280.80		103.64	0.3127	
50	87.87	280.80		103.64	0.3129	
60	87.91	280.80		103.64	0.3131	
70	88.08	280.80		103.64	0.3137	
80	88.15	280.80		103.64	0.3139	
90	88.35	280.80		103.64	0.3146	
100	88.44	280.80		103.64	0.3149	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
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10		154.98	-25.56	67.67		0.3777
20		196.56	-25.32	67.67		0.3742
30	86.19	280.80		103.64	0.3069	
40	86.42	280.80		103.64	0.3078	
50	86.87	280.80		103.64	0.3094	
60	87.22	280.80		103.64	0.3106	
70	87.65	280.80		103.64	0.3121	
80	87.92	280.80		103.64	0.3131	
90	88.28	280.80		103.64	0.3144	
100	88.42	280.80		103.64	0.3149	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.00	67.67		0.0887
20		115.92	-5.77	67.67		0.0852
30	62.85	165.60		103.64	0.3795	
40	63.08	165.60		103.64	0.3809	
50	63.52	165.60		103.64	0.3836	
60	63.88	165.60		103.64	0.3858	
70	64.31	165.60		103.64	0.3883	
80	64.58	165.60		103.64	0.3900	
90	64.94	165.60		103.64	0.3921	
100	65.07	165.60		103.64	0.3930	

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-5.12	67.67		0.0757
20		115.92	-4.91	67.67		0.0725
30	63.03	165.60		103.64	0.3806	
40	63.24	165.60		103.64	0.3819	
50	63.63	165.60		103.64	0.3843	
60	63.95	165.60		103.64	0.3862	
70	64.33	165.60		103.64	0.3884	
80	64.57	165.60		103.64	0.3899	
90	64.98	165.60		103.64	0.3924	
100	65.11	165.60		103.64	0.3932	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.00	67.67		0.0887
20		115.92	-5.77	67.67		0.0852
30	62.85	165.60		103.64	0.3795	
40	63.08	165.60		103.64	0.3809	
50	63.52	165.60		103.64	0.3836	
60	63.88	165.60		103.64	0.3858	
70	64.31	165.60		103.64	0.3883	
80	64.58	165.60		103.64	0.3900	
90	64.94	165.60		103.64	0.3921	
100	65.07	165.60		103.64	0.3930	

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.00	67.67		0.0887
20		115.92	-5.77	67.67		0.0852
30	62.85	165.60		103.64	0.3795	
40	63.08	165.60		103.64	0.3809	
50	63.52	165.60		103.64	0.3836	
60	63.88	165.60		103.64	0.3858	
70	64.31	165.60		103.64	0.3883	
80	64.58	165.60		103.64	0.3900	
90	64.94	165.60		103.64	0.3921	

100 65.07 165.60 103.64 0.3930

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.00	67.67		0.0887
20		115.92	-5.77	67.67		0.0852
30		165.60	-3.06	103.64		0.0295
40		165.60	-2.83	103.64		0.0273
50		165.60	-2.38	103.64		0.0230
60		165.60	-2.03	103.64		0.0195
70		165.60	-1.60	103.64		0.0154
80		165.60	-1.33	103.64		0.0128
90		165.60	-0.74	103.64		0.0072
100		165.60	-0.61	103.64		0.0059

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-5.75	56.40		0.1020
20		96.60	-5.53	56.40		0.0980
30	62.97	138.00		86.37	0.4563	
40	63.19	138.00		86.37	0.4579	
50	63.62	138.00		86.37	0.4610	
60	63.96	138.00		86.37	0.4635	
70	64.37	138.00		86.37	0.4664	
80	64.63	138.00		86.37	0.4684	
90	64.97	138.00		86.37	0.4708	
100	65.10	138.00		86.37	0.4717	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		75.60	-5.75	56.40		0.1020
20		96.60	-5.53	56.40		0.0980
30		138.00	-2.93	86.37		0.0340
40		138.00	-2.71	86.37		0.0314
50		138.00	-2.29	86.37		0.0265
60		138.00	-1.94	86.37		0.0225
70		138.00	-1.54	86.37		0.0178
80		138.00	-1.27	86.37		0.0148
90		138.00	-0.71	86.37		0.0083
100		138.00	-0.58	86.37		0.0067

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		90.72	-6.00	67.67		0.0887
20		115.92	-5.77	67.67		0.0852
30	62.85	165.60		103.64	0.3795	
40	63.08	165.60		103.64	0.3809	
50	63.52	165.60		103.64	0.3836	
60	63.88	165.60		103.64	0.3858	
70	64.31	165.60		103.64	0.3883	
80	64.58	165.60		103.64	0.3900	
90	64.94	165.60		103.64	0.3921	
100	65.07	165.60		103.64	0.3930	

Absolute Maximum of the all of the Stress Ratio's 0.4717

Governing Element: Tophead
 Governing Load Case 18 : FS+BS+IP+OW

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
---------------------	----------------------	----------------------------	-------------------------	---------------------------	-----------------	------------

Skirt-2	...	-5.5	0.3	...	...	+ Tens
Skirt-2	...	-5.5	-0.3	...	-5.8	- Comp
Skirt-1	...	-5.3	0.3	...	...	+ Tens
Skirt-1	...	-5.3	-0.3	...	-5.5	- Comp
Btm_head	65.6	-2.8	0.1	...	63.0	+ Tens
Btm_head	65.6	-2.8	-0.1	...	...	- Comp
Shell-1	65.6	-2.6	0.1	...	63.2	+ Tens
Shell-1	65.6	-2.6	-0.1	...	...	- Comp
Shell-2	65.6	-2.1	0.1	...	63.6	+ Tens
Shell-2	65.6	-2.1	-0.1	...	...	- Comp
Shell-3	65.6	-1.8	0.1	...	64.0	+ Tens
Shell-3	65.6	-1.8	-0.1	...	...	- Comp
Shell-4	65.6	-1.4	0.1	...	64.4	+ Tens
Shell-4	65.6	-1.4	-0.1	...	...	- Comp
Shell-5	65.6	-1.1	0.1	...	64.6	+ Tens
Shell-5	65.6	-1.1	-0.1	...	...	- Comp
Shell-6	65.6	-0.7	...	...	65.0	+ Tens
Shell-6	65.6	-0.7	...	...	...	- Comp
Top_head	65.6	-0.5	...	...	65.1	+ Tens
Top_head	65.6	-0.5	...	...	...	- Comp

Skirt Support Analysis Per EN 13445-3 Section 16.12:

The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-1

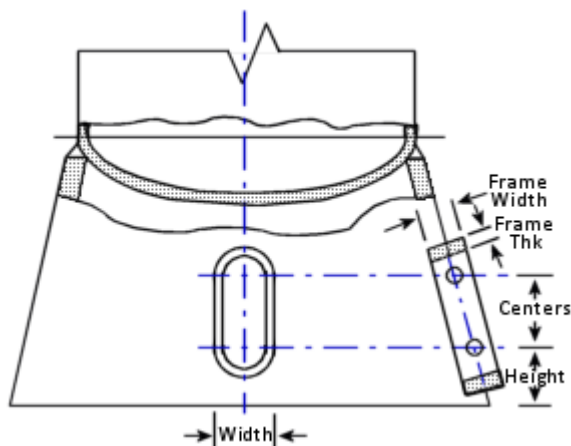
Total vessel (empty) weight	Fg	1168792.000 N
Head Weight below section 2-2	DeltaFg	91552.047 N
Weight of vessel contents (liquid)	Ff	0.000 N
Vessel Gross Weight	Fg + Ff	1168792.000 N
Skirt Mean Diameter	Dz	5527.500 mm
Skirt Analysis Thickness (Corroded)	ez	12.500 mm
Skirt Height	H	1500.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	3222282240.000 N-mm
Applied Moment - Bottom of Skirt	Mb	3625572352.000 N-mm
Weight at Bottom of Skirt	W	1193151.500 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Elliptical
Head Mean Diameter at Skirt Top	Db	5529.000 mm
Head Corroded Thickness	eb	23.000 mm

Per EN 13445 Section 7.5.4 [R, r]:

$$\begin{aligned}
 h_i &= 1378 \text{ mm} \\
 K &= D_i / (2 * h_i) = 5506 / (2 * 1378) = 2 \\
 R &= D_i (0.44 * K + 0.02) = 5506 * (0.44 * 2 + 0.02) = 4950 \text{ mm} \\
 r &= D_i ((0.5 / K) - 0.08) = 5506 * ((0.5 / 2) - 0.08) = 938 \text{ mm}
 \end{aligned}$$



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	829002416128.000 mm ⁴
Moment of Inertia	Iyy =	829002416128.000 mm ⁴
Section Modulus	W =	299955936.000 mm ³
Section Metal Area	A =	217064.438 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:

$$\begin{aligned}
 &= \sqrt{(\min(I_{xx}, I_{yy}) / \text{Area})} \\
 &= \sqrt{(\min(829.00E+09, 829.00E+09) / 217064)} \\
 &= 1954 \text{ mm}
 \end{aligned}$$

Skirt Slenderness Ratio [rg/L]:



$$= r_g / L$$

$$= 1954/1500$$

$$= 1.303$$

Correct the Moment [M4]:

top of skirt is the critical end
 Controlling Moment [M4] = 3.22E+09 N-mm

Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -3.22E+09/1199823744 - 1193151/217064$$

$$= -8.18 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +3.22E+09/1199823744 - 1193151/217064$$

$$= -2.81 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F_1 - F_g - F_f + 4 \cdot M_1 / D_z$$

$$= 0 - 1168792 - 0 + 4 \cdot 3.22E+09/5528$$

$$= 1162081 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F_1 - F_g - F_f - 4 \cdot M_1 / D_z$$

$$= 0 - 1168792 - 0 - 4 \cdot 3.22E+09/5528$$

$$= -3499665 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (F_{zp} + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (1162081 + 91552 + 0) / (\pi \cdot 5529 \cdot 23) + 1.1 \cdot 5529 / (4 \cdot 23)$$

$$= 69.25 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (F_{zq} + \Delta F_g + F_f) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (-3499665 + 91552 + 0) / (\pi \cdot 5529 \cdot 23) + 1.1 \cdot 5529 / (4 \cdot 23)$$

$$= 57.58 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (F_f + \Delta F_g) / (\pi \cdot D_b \cdot e_b) + P \cdot D_b / (4 \cdot e_b)$$

$$= (0 + 91552) / (\pi \cdot 5529 \cdot 23) + 1.1 \cdot 5529 / (4 \cdot 23)$$

$$= 66.34 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= F_{zp} / (\pi \cdot D_z \cdot e_z)$$

$$= 1162081 / (\pi \cdot 5528 \cdot 12.5)$$

$$= 5.35 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= F_{zq} / (\pi \cdot D_z \cdot e_z)$$

$$= -3499665 / (\pi \cdot 5528 \cdot 12.5)$$

$$= -16.12 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [$\cos(\gamma)$]:

$$= 1 - 0.5(D_b + e_b - D_z + e_z) / (r + e_b)$$

$$= 1 - 0.5(5529 + 23 - 5528 + 12.5) / (938 + 23)$$

$$= 0.98$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.98) = 11.3 \text{ deg}$$

$$\begin{aligned} \text{Dimension [a]:} \\ &= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))} \\ &= 0.5 \cdot \sqrt{(23^2 + 12.5^2 + 2 \cdot 23 \cdot 12.5 \cdot 0.98)} \\ &= 17.7 \text{ mm} \end{aligned}$$

$$\begin{aligned} \text{Bending moment at verification points p and q [Mp \& Mq]:} \\ M_p &= a \cdot F_{zp} = 17.67 \cdot 1162081.9 = 20.5444 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 17.67 \cdot -3499665.8 = -61.8706 \times 10^6 \text{ N-mm} \end{aligned}$$

$$\begin{aligned} \text{Correction factor [C]:} \\ &= 0.63 - 0.057 \left(\frac{eb}{ez} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{23}{12.5} \right)^2 \\ &= 0.437 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

$$\begin{aligned} \text{Stress } [\sigma_{1pb(o)} \& \sigma_{2pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot eb^2) \\ &= 0.44 \cdot 6 \cdot 20544446 / (\pi \cdot 5529 \cdot 23^2) \\ &= 5.86 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qb(o)} \& \sigma_{2qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot eb^2) \\ &= 0.44 \cdot 6 \cdot -61870600 / (\pi \cdot 5529 \cdot 23^2) \\ &= -17.65 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3pb(o)}]: \\ &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot ez^2) \\ &= 0.44 \cdot 6 \cdot 20544446 / (\pi \cdot 5528 \cdot 12.5^2) \\ &= 19.85 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{3qb(o)}]: \\ &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot ez^2) \\ &= 0.44 \cdot 6 \cdot -61870600 / (\pi \cdot 5528 \cdot 12.5^2) \\ &= -59.77 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Geometry variables, 16.12-32 to 36:} \\ eb/D_b &= 0.91/218 = 0.0042 \\ y &= 125 \cdot eb/D_b = 125 \cdot 23/5529 = 0.52 \end{aligned}$$

As this an elliptical head, it will be treated as a Korbbogen type head

$$\begin{aligned} \text{Geometric value [\alpha]:} \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot RYY + 3.994 \\ &= 1.51861 \cdot 2.71828 - 4.2335 \cdot 0.52 + 3.994 = 4.16 \end{aligned}$$

$$\begin{aligned} \text{Bending stress in sections 1-1 and 2-2 Equation 16.12-31 } [\sigma_{1b(p)} \& \sigma_{2b(p)}]: \\ &= (P + Ph) \cdot D_b \left(\left(\frac{\gamma}{\gamma_A} \right) \cdot \alpha - 1 \right) / (4 \cdot eb) \\ &= (1.1 + 0) \cdot 5529 \left(\left(\frac{11.3}{40} \right) \cdot 4.16 - 1 \right) / (4 \cdot 23) \\ &= 11.37 \text{ MPa} \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

$$\begin{aligned} \text{Stress } [\sigma_{1pitot}]: \\ &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 69.2 - 5.86 + 11.4 \\ &= 74.75 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1potot}]: \\ &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 69.2 + 5.86 - 11.4 \\ &= 63.74 \text{ MPa} \end{aligned}$$

$$\begin{aligned} \text{Stress } [\sigma_{1qitot}]: \\ &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \end{aligned}$$



$$= 57.6 - -17.7 + 11.4$$

$$= 86.59 \text{ MPa}$$

Stress [σ_{1qotot}]:

$$= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)}$$

$$= 57.6 + -17.7 - 11.4$$

$$= 28.56 \text{ MPa}$$

Stress [σ_{2pitot}]:

$$= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)}$$

$$= 66.3 + 5.86 + 11.4$$

$$= 83.57 \text{ MPa}$$

Stress [σ_{2potot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.3 - 5.86 - 11.4$$

$$= 49.11 \text{ MPa}$$

Stress [σ_{2qitot}]:

$$= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)}$$

$$= 66.3 + -17.7 + 11.4$$

$$= 60.05 \text{ MPa}$$

Stress [σ_{2qotot}]:

$$= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)}$$

$$= 66.3 - -17.7 - 11.4$$

$$= 72.62 \text{ MPa}$$

Stress [σ_{3pitot}]:

$$= \sigma_{3pm} - \sigma_{3pb}$$

$$= 5.35 - 19.8$$

$$= -14.49 \text{ MPa}$$

Stress [σ_{3potot}]:

$$= \sigma_{3pm} + \sigma_{3pb}$$

$$= 5.35 + 19.8$$

$$= 25.20 \text{ MPa}$$

Stress [σ_{3qitot}]:

$$= \sigma_{3qm} - \sigma_{3qb}$$

$$= -16.1 - -59.8$$

$$= 43.65 \text{ MPa}$$

Stress [σ_{3qotot}]:

$$= \sigma_{3qm} + \sigma_{3qb}$$

$$= -16.1 + -59.8$$

$$= -75.90 \text{ MPa}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (69.2/138)^2$$

$$= 390.83 \text{ MPa}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (57.6/138)^2$$

$$= 397.98 \text{ MPa}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (66.3/138)^2$$

$$= 392.74 \text{ MPa}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2$$

$$= 138 (3 - (1/1.5)) \cdot (66.3/138)^2$$

$$= 392.74 \text{ MPa}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3pm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (5.35/138)^2)$$

$$= 413.86 \text{ MPa}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:

$$= fs(3 - (1 / 1.5) \cdot (\sigma_{3qm} / f)^2)$$

$$= 138 (3 - (1/1.5) \cdot (-16.1/138)^2)$$

$$= 412.74 \text{ MPa}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	74.75	390.83	Pass	46
Stot1po	63.74	390.83	Pass	47
Stot1qi	86.59	397.98	Pass	48
Stot1qo	28.56	397.98	Pass	49
Stot2pi	83.57	392.74	Pass	50
Stot2po	49.11	392.74	Pass	51
Stot2qi	60.05	392.74	Pass	52
Stot2qo	72.62	392.74	Pass	53
Stot3pi	-14.49	413.86	Pass	54
Stot3po	25.20	413.86	Pass	55
Stot3qi	43.65	412.74	Pass	56
Stot3qo	-75.90	412.74	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	13863.423 mm
Center of Gravity of the Nozzles	9906.352 mm
Center of Gravity of the Added Weights (Operating)	17001.471 mm
Center of Gravity of the Added Weights (Empty)	14601.675 mm
Center of Gravity of Bare Shell New and Cold	10855.475 mm
Center of Gravity of Bare Shell Corroded	10835.240 mm
Vessel CG in the Operating Condition	12041.544 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	11662.283 mm
Vessel CG in the Test Condition	11769.984 mm

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	5540.0000	mm
Skirt Thickness	STHK	14.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.5000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		SA-285 C	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	28.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		SA-285 C	
Basing Ope. All. Stress	BASOPE, f4	108.00	MPa
Basing Yield Stress		207.00	MPa
Inside Diameter of Basing	DI, D4	5452.0000	mm
Outside Diameter of Basing	DOU, D5	5758.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	30.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	114.00	MPa
Number of Bolts	RN, nb	20	
Diameter of Bolt Circle	DC, DBC	5656.0000	mm
Bolt Allowable Shear Stress		138.000	MPa
Ultimate Comp. Strength of Concrete	FPC	20.7	MPa
Allowable Comp. Strength of Concrete	FC	8.3	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	12.0000	mm
Width of Gussets at Top Plate	TWDT	109.0000	mm
Width of Gussets at Base Plate	BWDT	109.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	207.0	MPa
Height of Chair	h1	245.0000	mm
Distance between Gussets	RG, b6	102.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	51.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	30.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	109.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	36.0000	mm
External Corrosion Allowance	CA	1.5000	mm
Dead Weight of Vessel	DW	1196261.5	N
Operating Weight of Vessel	ROW	1244330.5	N
Earthquake Moment on Basing	EQMOM	4244424448.0	N-mm
Wind Moment on Basing	WIMOM	3222676736.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:
 $= ((4 \cdot M/DC) - W) / nb$ per Jawad & Farr, Eq. 12.3
 $= ((4 \cdot 4.24E+09/5656) - 1190243) / 20$
 $= 90512.7 \text{ N}$

Bolt Load, Windload Empty Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 3.22E+09/5656) - 1196261) / 20$$

$$= 54096.8 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/5656) - 1244330) / 20$$

$$= -62216.5 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 390.7817 mm²
 Area Available in all the Bolts Abss * RN : 7815.6343 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.187	0.280	1.175	2.691	0.462	0.774	258802.2	1449045.8	49888.9
4	0.233	0.210	1.320	2.588	0.452	0.778	272784.2	1463028.0	54671.3
6	0.221	0.227	1.285	2.613	0.454	0.777	269258.6	1459502.2	53437.0
8	0.230	0.229	1.311	2.594	0.453	0.777	271900.9	1462144.6	54360.2
10	0.228	0.228	1.305	2.599	0.453	0.777	271239.2	1461482.9	54128.0
12	0.228	0.228	1.305	2.599	0.453	0.777	271294.4	1461538.0	54147.3

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 271294 / (0.44 \cdot 5656 \cdot 2.6)$$

$$= 83.9 \text{ MPa, Should be } \leq 114 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 20 \cdot 391 / (\pi \cdot 5656)$$

$$= 0.44 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc / [(2kd + t3) / (2kd)]$$

$$= 2.52 / [(2 \cdot 0.23 \cdot 5656 + 153) / (2 \cdot 0.23 \cdot 5656)]$$

$$= 2.67 \text{ MPa, Should be } \leq 8.27 \text{ MPa}$$

Values for table 10.3, l = 109, b = 444, l/b = 0.24537309

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(3029, 14197) = 14197 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow)} + Ca$$

$$= \sqrt{(6 \cdot 14197 / 137)} + 1.5$$

$$= 26.5 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 109, b = 102, b/l = 0.93577987

As b/l (0.94) is less than 1, inverting b/l = 1.07 0.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3 (\ln((2 \cdot l \sin(\pi \cdot a / l)) / (\pi \cdot g))) + 1] - [(0.7 - g2)P / (4 \cdot \pi)]$$

$$= 44545 / (4 \cdot \pi) [1.3 (\ln((2 \cdot 109 \cdot \sin(\pi \cdot 58 / 109))) / (\pi \cdot 25))) + 1] - [(0.7 - 0.13) \cdot 44545 / (4 \cdot \pi)]$$

$$= 6199.4932 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb)} + Ca$$

$$= \sqrt{(6 \cdot \text{Abs}(6199) / 138)} + 1.5$$

$$= 17.9186 \text{ mm}$$

Required Gusset Plate Thickness [tg]:
 $= P / (\text{Stress Term} \cdot l_{avg}) + C_a$
 $= 44545 / (124 \cdot 109) + 1.5$
 $= 11 \text{ (not less than } 9.52 + 1.5 \text{) mm}$

Bolt spacing [m]:
 $= \pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
 $= \pi \cdot 5656 / 20 = 888 \text{ mm}$

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
 $= 1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$
 $= 1.76(44545 \cdot 58 / (888 \cdot 215 \cdot 162))^{2/3} \cdot 27701^{1/3} + C_a$
 $= 4.72 + 1.5 = 6.22 \text{ mm}$

Shear Stress in a Single Bolt [tb]:
 $= \text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 270434 / (391 \cdot 20)$
 $= 34.6 \text{ MPa. Must be less than } 138.0 \text{ MPa.}$

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	26.4708	mm
Actual Basing Thickness as entered by user	28.0000	mm
Required Thickness of Chair Cap	17.9186	mm
Actual Top Ring Thickness as entered by user	30.0000	mm
Required Gusset thickness, + CA	11.0250	mm
Actual Gusset Thickness as entered by user	12.0000	mm
Required Thickness of Skirt for Local Stress	6.2220	mm
Given Thickness of Skirt	14.0000	mm
Required Gusset Height to meet local stress	21.9190	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
 $= \text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$
 $= 29 \cdot (14 - 1.5)$
 $= 363 \text{ N/mm}$

Results for Computed Minimum Basing Weld Size [BWeld]:
 $= W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 363 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 3.35 \text{ mm}$

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
 $= \text{Bolt Load} / (C_{mwh} + 2 \cdot (H_g + T_{ta}))$
 $= 44545 / (176 + 2 \cdot (187 + 30))$
 $= 73 \text{ N/mm}$

Horizontal Plate Load [Wh]:
 $= \text{Bolt Load} \cdot e / (C_{mwh} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2)$
 $= 44545 \cdot 58 / (176 \cdot (217) + 0.6667 \cdot (217)^2)$
 $= 37.1 \text{ N/mm}$

Resultant Weld Load [Wr]:
 $= \sqrt{ W_v^2 + W_h^2 }$
 $= \sqrt{ 73^2 + 37.1^2 }$
 $= 81.9 \text{ N/mm}$

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
 $= W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
 $= 81.9 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$
 $= 0.76 \text{ mm}$

Results for Computed Minimum Gusset to Top Plate Weld Size



Weld Load [Wv]:

$$= \text{Bolt Load} / (2 \cdot \text{TopWth})$$

$$= 44545 / (2 \cdot 109)$$

$$= 204 \text{ N/mm}$$

Weld Load [Wh]:

$$= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth})$$

$$= 44545 \cdot 58 / (2 \cdot 217 \cdot 109)$$

$$= 54.6 \text{ N/mm}$$

Resultant Weld Load [Wr]:

$$= \sqrt{Wv^2 + Wh^2}$$

$$= \sqrt{204^2 + 54.6^2}$$

$$= 212 \text{ N/mm}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$$

$$= 212 / [(0.4 \cdot 192) \cdot 2 \cdot 0.707]$$

$$= 1.95 \text{ mm}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	6.3500	mm
Required Gusset to Skirt Double Fillet Weld Size	4.7625	mm
Required Top Plate to Skirt Weld Size	6.3500	mm
Required Gusset to Top Plate Double Fillet Weld Size	1.9522	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N33"

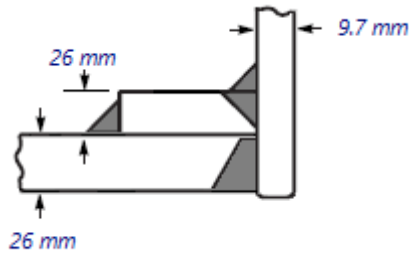
From: 30

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	5500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		3.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		160	
Flange Material		SA-105	+N
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	130.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		20.5500	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N33"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.734 in.
Actual Thickness Used in Calculation	0.383 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 37.7 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.3688 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	150.8618 mm
Parallel to Vessel Wall, opening length	d	75.4309 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.8364 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

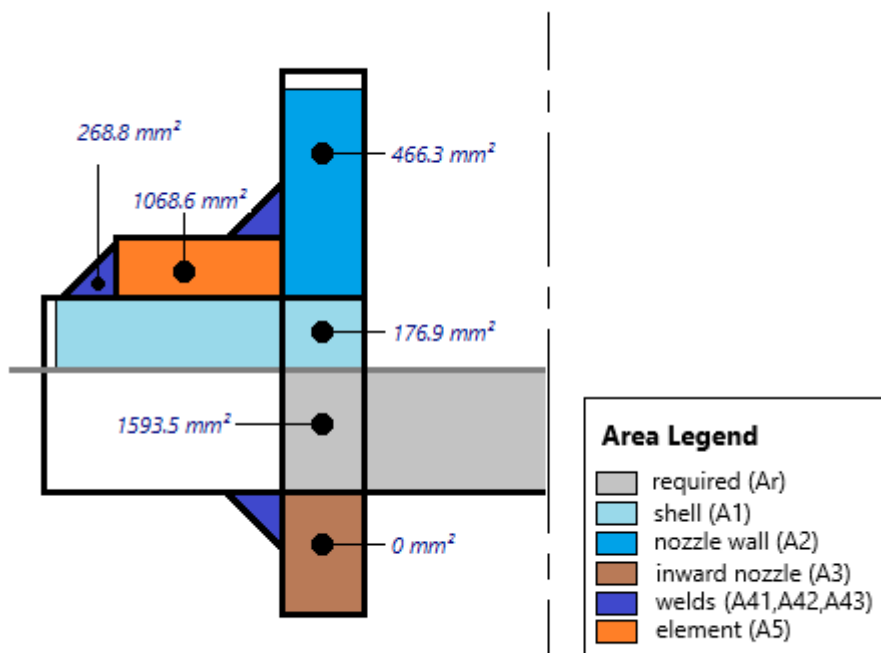


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1593.509	NA	NA
Area in Shell	A1	176.903	NA	NA
Area in Nozzle Wall	A2	466.322	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	268.767	NA	NA
Area in Element	A5	1068.600	NA	NA
TOTAL AREA AVAILABLE	Atot	1980.592	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	115.1122	26.0000 mm
Based on given Pad Diameter:	130.0000	16.5819 mm
Based on the Estimated Diameter Limit:	149.2743	11.2882 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (75.4 \cdot 20.6 \cdot 1 + 2 \cdot 6.73 \cdot 20.6 \cdot 1 \cdot (1 - 0.86))$
 $= 1593.509 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 75.4(1 \cdot 23 - 1 \cdot 20.6) - 2 \cdot 6.73$
 $(1 \cdot 23 - 1 \cdot 20.6) \cdot (1 - 0.86)$
 $= 176.903 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.8) \cdot (6.73 - 0.37) \cdot 0.86$

= 466.322 mm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
 = (100) • 0.86 + (0) • 0.86 + 183² • 1
 = 268.767 mm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (130 - 88.9) • 26 • 1
 = 1068.600 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.3688 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.8711 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.8711 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 7.8000 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[7.8, max(25.9, 4.5)]
 = 7.8000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.37, 7.8)
 = 7.8000 mm

Available Nozzle Neck Thickness = 9.7345 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.7,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	259.3 MPa	Passed
Occasional	:	2.9,	Allowable	:	156.9 MPa	Passed
Shear	:	25.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.73, tr = 0.37, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.055, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.73, tr = 0.37, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.055, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.73, tr = 0.37, c = 3 mm, E* = 1
 Thickness Ratio = tr * E*/(tg - c) = 0.055, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 26, tr = 20.6, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -40 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.73, tr = 0.37, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.055$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -40 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -40 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15/5.10 = 0.225$

Weld Size Calculations, Description: N33"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$9.8980 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (1594 - 177 + 2 \cdot 6.73 \cdot 0.86 \cdot (1 \cdot 23 - 20.6)) \cdot 138)$
 $= 199300.98 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (466 + 1069 + 269 - 0 \cdot 0.86) \cdot 138$
 $= 248887.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (466 + 0 + 85.5 + (265)) \cdot 138$
 $= 112697.12 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (466 + 0 + 269 + 1069 + (265)) \cdot 138$
 $= 285439.31 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 88.9 \cdot 10 \cdot 0.49 \cdot 118$
 $= 80734. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 130 \cdot 14 \cdot 0.49 \cdot 138$
 $= 193299. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 41.1) \cdot (9.73 - 3) \cdot 0.7 \cdot 118$
 $= 71788. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 88.9 \cdot 26 \cdot 0.74 \cdot 138$
 $= 370740. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 88.9 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 327962. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (193299 + 71788) = 265087 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (80734 + 370740 + 327962 + 0) = 779435 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (193299 + 327962 + 0) = 521261 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 265087 N , must exceed W = 199300 N or W1 = 248887 N
 Path 2-2 = 779435 N , must exceed W = 199300 N or W2 = 112697 N
 Path 3-3 = 521260 N , must exceed W = 199300 N or W3 = 285439 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.1988 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.1988 mm

Input Echo, WRC107/537 Item 1, Description: N33" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	9899.999	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow

Diameter Basis for Nozzle	Nbasis	ID
Corrosion Allowance for Nozzle	Can	3.0000 mm

Nozzle Diameter	Dn	69.431	mm
Nozzle Thickness	Tn	9.735	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	26.000	mm
Diameter of Reinforcing Pad	Dpad	130.000	mm
Design Internal Pressure	Dp	1.147	MPa
Include Pressure Thrust		No	

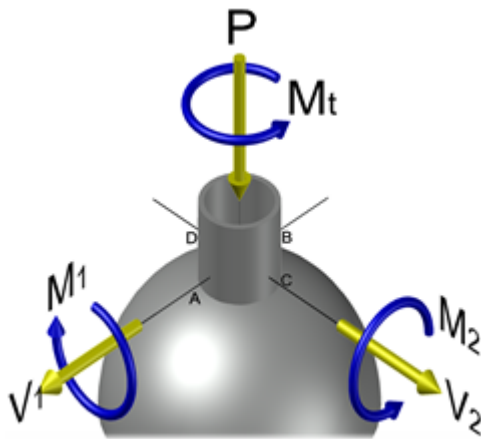
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-4799.8	N
Longitudinal Shear (SUS)	(V1) V1	4242.4	N
Circumferential Shear (SUS)	(Vc) V2	4242.4	N
Circumferential Moment (SUS)	(Mc) M1	835246.2	N-mm
Longitudinal Moment (SUS)	(Ml) M2	835246.2	N-mm
Torsional Moment (SUS)	Mt	1080379.9	N-mm

Use Interactive Control No
 WRC107 Version Version March 1979

Include Pressure Stress Indices per Div. 2 No
 Compute Pressure Stress per WRC-368 No
 Local Loads applied at end of Nozzle/Attachment No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1
 Warning - The ratio Ro/Rm, 0.0090, must be > 0.0571 and < 0.571.

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 88.9 + 2 \cdot 1.65 \cdot \sqrt{(4964.5 (26.0 - 3.0))}$$

$$= 1204.005 \text{ mm}$$

WRC 107 Stress Calculation for SUStained loads:

Radial Load	P	-4799.8	N
Circumferential Shear	(VC) V2	4242.4	N
Longitudinal Shear	(VL) V1	4242.4	N
Circumferential Moment	(MC) M1	835246.2	N-mm
Longitudinal Moment	(ML) M2	835246.2	N-mm
Torsional Moment	MT	1080379.9	N-mm

Dimensionless Param: U = 0.09 TAU = 6.10 RHO = 4.00 (7.28)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 4	0.11313	(A,B,C,D)
M(x) / P	SP 4	0.04621	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 4	0.58630	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 4	0.26698	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 4	0.58630	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 4	0.26698	(A,B,C,D)
N(y) * T / P	SP 4	0.32355	(A,B,C,D)
M(y) / P	SP 4	0.34564	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 4	0.10306	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 4	2.18122	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 4	0.10306	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 4	2.18122	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Rad.	Bend. P	0.6	-0.6	0.6	-0.6	0.6	-0.6	0.6	-0.6
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-0.4	-0.4	0.4	0.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-1.1	1.1	1.1	-1.1
Rad.	Memb. ML	-0.4	-0.4	0.4	0.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-1.1	1.1	1.1	-1.1	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-0.8	0.4	2.3	-1.0	-0.8	0.4	2.3	-1.0
Tang.	Memb. P	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang.	Bend. P	4.1	-4.1	4.1	-4.1	4.1	-4.1	4.1	-4.1
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-0.1	-0.1	0.1	0.1
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-9.2	9.2	9.2	-9.2
Tang.	Memb. ML	-0.1	-0.1	0.1	0.1	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-9.2	9.2	9.2	-9.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-4.5	5.6	14.1	-12.6	-4.5	5.6	14.1	-12.6
Shear VC		0.6	0.6	-0.6	-0.6	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.6	-0.6	0.6	0.6
Shear MT		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tot. Shear		2.4	2.4	1.2	1.2	1.2	1.2	2.4	2.4
Str. Int.		6.1	7.1	14.2	12.8	4.8	5.9	14.6	13.1

Unitless Prm: U = 0.19 TAU = 0.00 (9.15) RHO = 0.00 (3.42)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.23358	(A,B,C,D)
M(x) / P	SR 2	0.20273	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15441	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.81020	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15441	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.81020	(A,B,C,D)
N(y) * T / P	SR 2	0.07016	(A,B,C,D)
M(y) / P	SR 2	0.05931	(A,B,C,D)

$N(y) * T * \sqrt{R_m * T} / MC$ SR 3 0.04714 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / MC$ SR 3 0.24072 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SR 3 0.04714 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SR 3 0.24072 (A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Rad. Bend. P		11.0	-11.0	11.0	-11.0	11.0	-11.0	11.0	-11.0
Rad. Memb. MC		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Rad. Bend. MC		0.0	0.0	0.0	0.0	-22.7	22.7	22.7	-22.7
Rad. Memb. ML		-0.7	-0.7	0.7	0.7	0.0	0.0	0.0	0.0
Rad. Bend. ML		-22.7	22.7	22.7	-22.7	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-10.3	13.1	36.6	-30.9	-10.3	13.1	36.6	-30.9
Tang. Memb. P		0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.6
Tang. Bend. P		3.2	-3.2	3.2	-3.2	3.2	-3.2	3.2	-3.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-6.7	6.7	6.7	-6.7
Tang. Memb. ML		-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-6.7	6.7	6.7	-6.7	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-3.1	3.9	10.8	-9.1	-3.1	3.9	10.8	-9.1
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tot. Shear		2.7	2.7	0.9	0.9	0.9	0.9	2.7	2.7
Str. Int.		11.2	13.8	36.6	30.9	10.4	13.1	36.9	31.2

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Rad. Pl (SUS)		-0.2	-0.2	0.6	0.6	-0.2	-0.2	0.6	0.6
Rad. Q (SUS)		-0.6	0.6	1.7	-1.7	-0.6	0.6	1.7	-1.7
Long. Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Long. Pl (SUS)		0.6	0.6	0.7	0.7	0.6	0.6	0.7	0.7
Long. Q (SUS)		-5.1	5.1	13.4	-13.4	-5.1	5.1	13.4	-13.4
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.6	0.6	-0.6	-0.6	-0.6	-0.6	0.6	0.6
Shear Q (SUS)		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Pm+Pl (SUS)		58.9	58.9	59.3	59.3	58.9	58.9	59.3	59.3
Pm+Pl+Q (Total)		58.4	64.6	72.2	57.1	57.6	63.9	72.5	57.4

Vessel Stress Summation Comparison (MPa):

Type of	Max. S.I.	S.I. Allowable	Result
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Stress Int.			
Pm (SUS)	57.98	138.00	Passed
Pm+Pl (SUS)	59.28	207.00	Passed
Pm+Pl+Q (TOTAL)	72.53	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Rad.	Pl (SUS)	1.4	1.4	2.8	2.8	1.4	1.4	2.8	2.8
Rad.	Q (SUS)	-11.7	11.7	33.7	-33.7	-11.7	11.7	33.7	-33.7
Long.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Long.	Pl (SUS)	0.4	0.4	0.9	0.9	0.4	0.4	0.9	0.9
Long.	Q (SUS)	-3.5	3.5	10.0	-10.0	-3.5	3.5	10.0	-10.0
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear	Q (SUS)	1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Pm+Pl (SUS)		125.5	125.5	126.7	126.7	125.5	125.5	126.7	126.7
Pm+Pl+Q (Total)		121.3	137.3	160.1	114.4	120.5	136.7	160.4	114.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.52	138.00	Passed
Pm+Pl (SUS)	126.71	207.00	Passed
Pm+Pl+Q (TOTAL)	160.38	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N9B2"

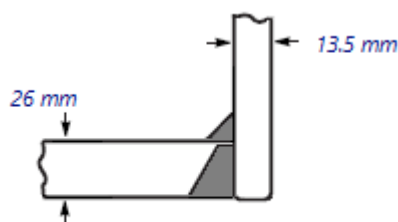
From: 40

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N9B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tt	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N9B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000$ mm

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N9B2"

Intermediate Calc. for nozzle/shell welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

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Input, Nozzle Desc: N10B2"

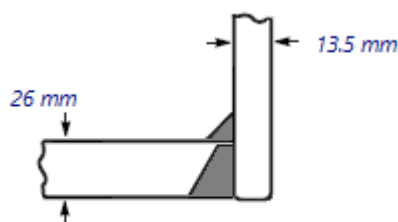
From: 40

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		250.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N10B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tr	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10B2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N10B2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 61.86 mm

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Input, Nozzle Desc: M1B24"

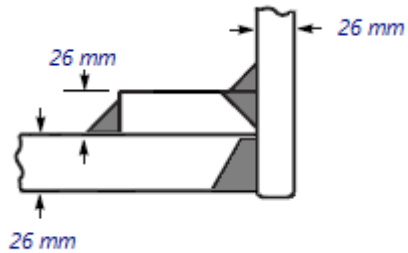
From: 40

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1500.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+N
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1B24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.024 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.15 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1.15)$
 $= 2.5254 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1127.2000 mm
Parallel to Vessel Wall, opening length	d	563.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		57.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

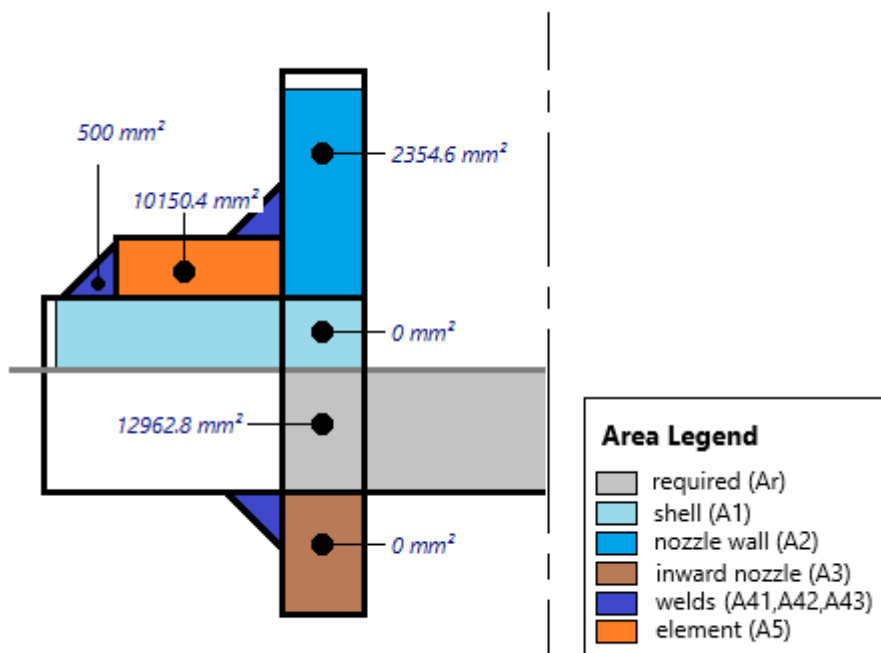


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12962.799	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	2354.583	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	500.000	NA	NA
Area in Element	A5	10150.399	NA	NA
TOTAL AREA AVAILABLE	Atot	13004.984	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 998.3774 26.0000 mm
 Based on given Pad Diameter: 1000.0000 25.8919 mm
 Based on Shell or Nozzle Thickness: 998.3774 26.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (564 \cdot 23 \cdot 1 + 2 \cdot 23 \cdot 23 \cdot 1 \cdot (1 - 1))$
 $= 12962.799 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 564(1 \cdot 23 - 1 \cdot 23) - 2 \cdot 23$
 $(1 \cdot 23 - 1 \cdot 23) \cdot (1 - 1)$
 $= 0.001 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 57.5) \cdot (23 - 2.53) \cdot 1$

= 2354.583 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 20^2 \cdot 1$
 = 500.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1000 - 610) \cdot 26 \cdot 1$
 = 10150.399 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.5254$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 5.5254 mm

Available Nozzle Neck Thickness = 26.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 23$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 26$, $t_r = 23$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Gov. MDMT of the Nozzle : -34 °C
 Gov. MDMT of the Reinforcement Pad : -34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.15/1.96 = 0.585

Weld Size Calculations, Description: M1B24"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm
 Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{min} Pad	14.1400 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12963 - 0.00085 + 2 \cdot 23 \cdot 1 \cdot (1 \cdot 23 - 23)) \cdot 138)$
 $= 1788713.88 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_2) \cdot S_v$
 $= (2355 + 10150 + 500 - 0 \cdot 1) \cdot 138$
 $= 1794535.00 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_1)) \cdot S_v$
 $= (2355 + 0 + 100 + (1058)) \cdot 138$
 $= 484695.22 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_1)) \cdot S$
 $= (2355 + 0 + 500 + 10150 + (1058)) \cdot 138$
 $= 1940526.62 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1000 \cdot 20 \cdot 0.49 \cdot 138$
 $= 2124164. \text{ N}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 293) \cdot (26 - 3) \cdot 0.7 \cdot 138$
 $= 2047057. \text{ N}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 26 \cdot 0.74 \cdot 138$
 $= 2542214. \text{ N}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 2248881. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SPEW} + \text{SNW}) = (2124164 + 2047057) = 4171221 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (647445 + 2542213 + 2248881 + 0) = 5438540 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2124164 + 2248881 + 0) = 4373045 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4171221 N , must exceed W = 1788713 N or W1 = 1794535 N
Path 2-2 = 5438540 N , must exceed W = 1788713 N or W2 = 484695 N
Path 3-3 = 4373045 N , must exceed W = 1788713 N or W3 = 1940526 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.9435 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.9435 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.455 %

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Input, Nozzle Desc: N9A2"

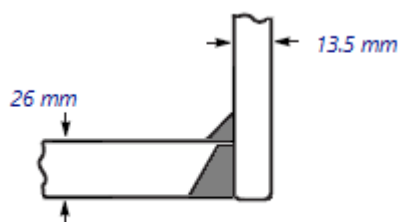
From: 60

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7500.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N9A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tt	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N9A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N9A2"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

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Input, Nozzle Desc: N10A2"

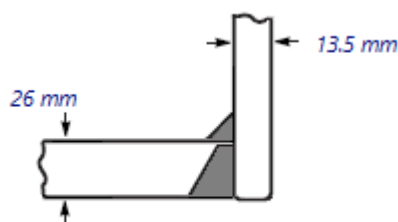
From: 60

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		7000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N10A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tt	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10A2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N10A2"

Intermediate Calc. for nozzle/shell welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

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Input, Nozzle Desc: M1A24"

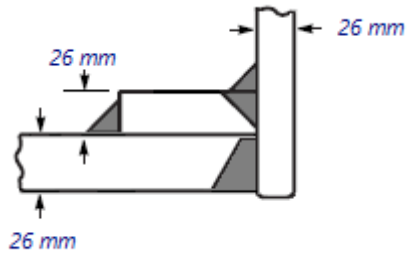
From: 60

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		8000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	+N
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	26.0000	mm
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1000.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		195.2000	mm
This is a Manway, Coupling or Access Opening.			
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: M1A24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	1.024 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.15 \cdot 305 / (138 \cdot 1 + 0.4 \cdot 1.15)$
 $= 2.5254 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1127.2000 mm
Parallel to Vessel Wall, opening length	d	563.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		57.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1, 1)$
 $= 1.000$

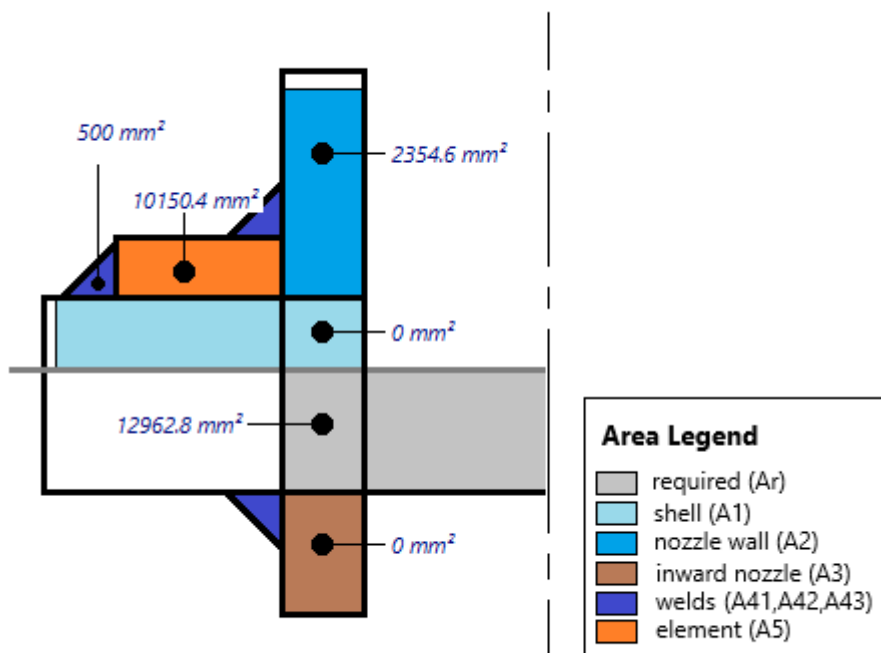


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12962.799	NA	NA
Area in Shell	A1	0.001	NA	NA
Area in Nozzle Wall	A2	2354.583	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	500.000	NA	NA
Area in Element	A5	10150.399	NA	NA
TOTAL AREA AVAILABLE	Atot	13004.984	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 998.3774 26.0000 mm
 Based on given Pad Diameter: 1000.0000 25.8919 mm
 Based on Shell or Nozzle Thickness: 998.3774 26.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (564 \cdot 23 \cdot 1 + 2 \cdot 23 \cdot 23 \cdot 1 \cdot (1 - 1))$
 $= 12962.799 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 564(1 \cdot 23 - 1 \cdot 23) - 2 \cdot 23$
 $(1 \cdot 23 - 1 \cdot 23) \cdot (1 - 1)$
 $= 0.001 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 57.5) \cdot (23 - 2.53) \cdot 1$

= 2354.583 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 1 + (0)^2 \cdot 1 + 20^2 \cdot 1$
 = 500.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(1000 - 610) \cdot 26 \cdot 1$
 = 10150.399 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 5.5254$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= t_a
 = 5.5254 mm

Available Nozzle Neck Thickness = 26.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:D, Curve:A)

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 2.53$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, $t_g = 26$, $t_r = 23$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: D

Govrn. thk, $t_g = 26$, $t_r = 23$, $c = 3$ mm, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Gov. MDMT of the Nozzle : -34 °C
 Gov. MDMT of the Reinforcement Pad : -34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C



Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 1.15/1.96 = 0.585

Weld Size Calculations, Description: M1A24"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm
 Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{min} Pad	14.1400 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_1 \cdot (E_1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (12963 - 0.00085 + 2 \cdot 23 \cdot 1 \cdot (1 \cdot 23 - 23)) \cdot 138)$
 $= 1788713.88 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - \text{Can} / .707)^2 \cdot f_2) \cdot S_v$
 $= (2355 + 10150 + 500 - 0 \cdot 1) \cdot 138$
 $= 1794535.00 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_1)) \cdot S_v$
 $= (2355 + 0 + 100 + (1058)) \cdot 138$
 $= 484695.22 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_1)) \cdot S$
 $= (2355 + 0 + 500 + 10150 + (1058)) \cdot 138$
 $= 1940526.62 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 610 \cdot 10 \cdot 0.49 \cdot 138$
 $= 647445. \text{ N}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.14/2.0) \cdot 1000 \cdot 20 \cdot 0.49 \cdot 138$
 $= 2124164. \text{ N}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 293) \cdot (26 - 3) \cdot 0.7 \cdot 138$
 $= 2047057. \text{ N}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.14/2) \cdot 610 \cdot 26 \cdot 0.74 \cdot 138$
 $= 2542214. \text{ N}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 610 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 2248881. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:



$$= (\text{SPEW} + \text{SNW}) = (2124164 + 2047057) = 4171221 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (647445 + 2542213 + 2248881 + 0) = 5438540 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2124164 + 2248881 + 0) = 4373045 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4171221 N , must exceed W = 1788713 N or W1 = 1794535 N
Path 2-2 = 5438540 N , must exceed W = 1788713 N or W2 = 484695 N
Path 3-3 = 4373045 N , must exceed W = 1788713 N or W3 = 1940526 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 16.9435 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 342.9435 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 4.455 %

This nozzle or its diameter limit intersects weld seam # [4] on this element. The distance between the nozzle and seam is, 505.17 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 563.60 mm

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Input, Nozzle Desc: N110"

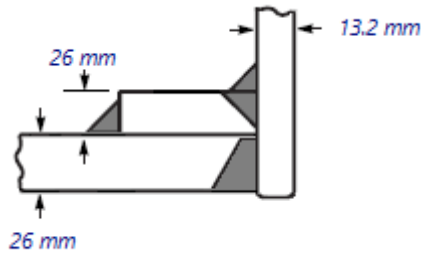
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Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	+N
Flange Type		Weld Neck	Flange
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	460.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		93.4750	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N110"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.710 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 126 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 1.2353 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934	mm
Parallel to Vessel Wall, opening length	d	252.6467	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		51.5041	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118 / 138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138 / 138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

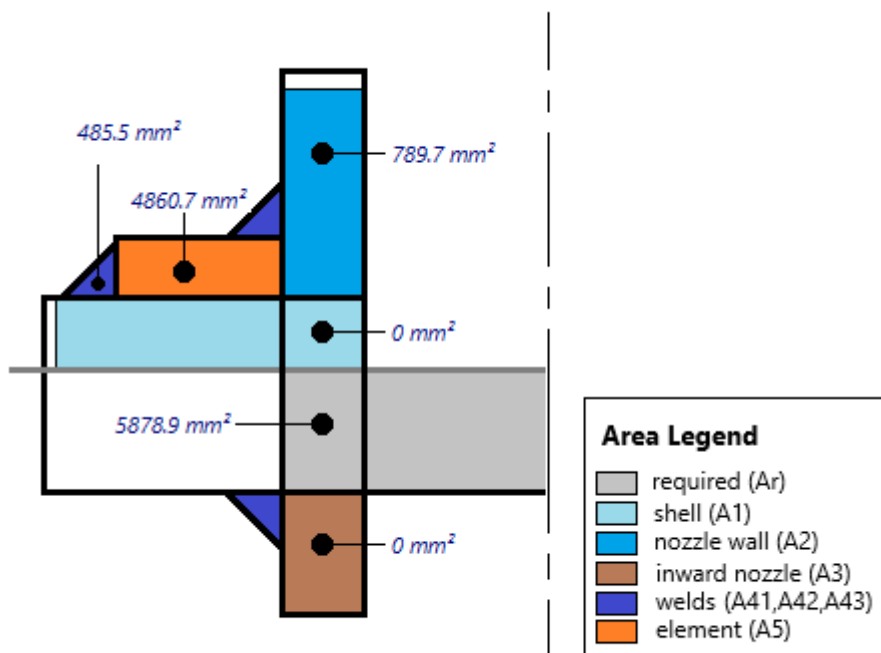


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5878.892	NA	NA
Area in Shell	A1	0.000	NA	NA
Area in Nozzle Wall	A2	789.736	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	485.506	NA	NA
Area in Element	A5	4860.700	NA	NA
TOTAL AREA AVAILABLE	Atot	6135.942	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	450.1135	26.0000 mm
Based on given Pad Diameter:	460.0000	24.6250 mm
Based on the Estimated Diameter Limit:	503.7059	19.9590 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (253 \cdot 23 \cdot 1 + 2 \cdot 10.2 \cdot 23 \cdot 1 \cdot (1 - 0.86))$
 $= 5878.892 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 253(1 \cdot 23 - 1 \cdot 23) - 2 \cdot 10.2$
 $(1 \cdot 23 - 1 \cdot 23) \cdot (1 - 0.86)$
 $= 0.000 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 51.5) \cdot (10.2 - 1.24) \cdot 0.86$

= 789.736 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_n / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1$
 = 485.506 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(460 - 273) \cdot 26 \cdot 1$
 = 4860.700 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2353 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 26.0000 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 26.0000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[11.1, max(26, 4.5)]
 = 11.1026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.24, 11.1)
 = 11.1026 mm

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained :	29.5,	Allowable :	118.0 MPa	Passed
Expansion :	0.0,	Allowable :	265.5 MPa	Passed
Occasional :	6.8,	Allowable :	156.9 MPa	Passed
Shear :	16.7,	Allowable :	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 26, tr = 23, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 1$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve D -34 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -34 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -34 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15/1.96 = 0.585$

Weld Size Calculations, Description: N110"

Intermediate Calc. for nozzle/shell Welds Tmin 10.2016 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	14.1400 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5879 - 0.00038 + 2 \cdot 10.2 \cdot 0.86 \cdot (1 \cdot 23 - 23)) \cdot 138)$
 $= 811217.94 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (790 + 4861 + 486 - 0 \cdot 0.86) \cdot 138$
 $= 846687.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (790 + 0 + 85.5 + (401)) \cdot 138$
 $= 176141.89 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (790 + 0 + 486 + 4861 + (401)) \cdot 138$
 $= 902056.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 118$
 $= 247968. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 460 \cdot 20 \cdot 0.49 \cdot 138$
 $= 977116. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 131) \cdot (13.2 - 3) \cdot 0.7 \cdot 118$
 $= 347881. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 273 \cdot 26 \cdot 0.74 \cdot 138$
 $= 1138700. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 273 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 1007311. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (977116 + 347881) = 1324996 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (247968 + 1138699 + 1007311 + 0) = 2393979 \text{ N}$

Failure Path [Path 3-3]:
 $= (Spew + Tngw + Sinw)$
 $= (977116 + 1007311 + 0) = 1984426 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1324996 N	, must exceed W = 811217 N	or W1 = 846687 N
Path 2-2 = 2393979 N	, must exceed W = 811217 N	or W2 = 176141 N
Path 3-3 = 1984426 N	, must exceed W = 811217 N	or W3 = 902056 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.3911 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.3910 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 252.65 mm

Input Echo, WRC297 Item 1, Description: N110" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	5500.000	mm
Shell Thickness	26.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 MPa
Vessel Hot S.I. Allowable	Smh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	246.647	mm

Nozzle Thickness	13.2016	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	460.000	mm
Thickness of Reinforcing Pad	26.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	1.15	MPa
Radial Load	P	-15999.28	N
Circumferential Shear	Vc	11999.50	N
Longitudinal Shear	Vl	15999.28	N
Circumferential Moment	Mc	0.800E+07	N-mm
Longitudinal Moment	Ml	0.104E+08	N-mm
Torsional Moment	Mt	0.120E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(2777.5(26.0 - 3.0))}$
 $= 1107.125 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5555.000	mm
Nozzle Outside Diameter	(d) =	273.050	mm
Vessel Thickness used	(T) =	49.000	mm
Nozzle Thickness used	(t) =	10.202	mm
T / t	=	4.803	
d / t	=	26.765	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.523$$

Nr/P	=	0.095
Mr/P	=	0.079
M0/P	=	0.177
N0/P	=	0.342
MrD/Mc	=	0.106
NrDL/Mc	=	0.113
M0d/Mc	=	0.292
N0DL/Mc	=	0.081
MrD/Ml	=	0.106
NrDL/Ml	=	0.112
M0D/Ml	=	0.269
N0DL/Ml	=	0.218

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3



Outplane Bending (ML)	-25	25	25	-25
Normal Pressure Stress	63	65	63	65

Outplane Stress Summary	44	82	102	38
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-10	10	10	-10
Inplane Pressure Stress	31	31	31	31

Inplane Stress Summary	23	37	47	21
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to	Top	Top	Bottom	Bottom
longitudinal plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	2	2	1	1
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	44	82	102	38
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	18	47	63	6
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-7	7	7	-7
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65

Inplane Stress Summary	58	68	76	56
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	58	68	77	56

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17
Hoop Stress Summary	15	16	22	23

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-19	-19	19	19
Axial Bending (ML)	-108	108	108	-108
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	-89	70	167	-108

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10
Shear Stress Summary	12	12	7	7

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	107	73	167	132

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	17	19	19	21
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	28	-28	28	-28
Axial Membrane (Mc)	-14	-14	14	14
Axial Bending (Mc)	-83	83	83	-83
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-60	50	137	-87
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	10	10	10	10

Shear Stress Summary	6	6	13	13
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	78	51	139	111
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(2764.5(26.0 - 3.0))}$$

$$= 1105.170 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	5529.000 mm
Nozzle Outside Diameter	(d) =	460.000 mm
Vessel Thickness used	(T) =	23.000 mm
Nozzle Thickness used	(t) =	103.677 mm
T / t	=	0.222
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.290$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.093
Mr/P	=	0.195
M0/P	=	0.057
N0/P	=	0.103
MrD/Mc	=	0.306
NrDL/Mc	=	0.079
M0d/Mc	=	0.093
N0DL/Mc	=	0.058
MrD/M1	=	0.161
NrDL/M1	=	0.096
M0D/M1	=	0.050



NØDL/M1 = 0.114

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-12	12	12	-12
Normal Pressure Stress	136	137	136	137
Outplane Stress Summary	132	138	167	122

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-41	41	41	-41
Inplane Pressure Stress	68	68	68	68
Inplane Stress Summary	61	73	152	-1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	132	138	167	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-1	-1	1	1
Outplane Bending (Mc)	-18	18	18	-18
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	68	68	68	68
Outplane Stress Summary	61	77	102	44

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	35	-35	35	-35
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-60	60	60	-60
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	136	137	136	137

Inplane Stress Summary	112	163	237	47
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	112	163	237	49

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	-1	-1	5	5	-1	-1	1	1
Circ. Q (SUS)	-18	18	32	-32	-4	4	10	-10
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-1	-1	1	1	2	2	2	2
Long. Q (SUS)	-7	7	13	-13	-14	14	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+Pl (SUS)	62.0	64.0	68.0	70.0	62.0	64.0	64.0	66.0
Pm+Pl+Q (Total)	44.2	82.1	100.1	38.2	58.1	68.2	74.3	56.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed
Pm+Pl+Q (TOTAL)	100.07	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	136	137	136	137	136	137	136	137
Circ. Pl (SUS)	-1	-1	7	7	0	0	4	4
Circ. Q (SUS)	-2	2	22	-22	-25	25	95	-95
Long. Pm (SUS)	68	68	68	68	68	68	68	68
Long. Pl (SUS)	-2	-2	6	6	2	2	4	4
Long. Q (SUS)	-6	6	76	-76	-8	8	28	-28
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	136.0	137.0	136.0	137.0	136.0	137.0	136.0	137.0
Pm+Pl (SUS)	135.0	136.0	143.0	144.0	136.0	137.0	140.0	141.0
Pm+Pl+Q (Total)	133.0	138.0	165.1	124.0	111.0	162.0	235.0	46.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	137.00	138.00	Passed
Pm+Pl (SUS)	144.00	207.00	Passed
Pm+Pl+Q (TOTAL)	235.01	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-1	-1	5	5	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-18	-18	20	20	-13	-13	15	15
Long. Q (SUS)	-80	80	136	-136	-55	55	111	-111
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-3	-3	3	3
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	25.3	26.3	28.5	28.6	23.8	24.7	24.4	24.6
Pm+Pl+Q (Total)	107.7	72.5	164.4	131.0	79.2	51.5	135.4	110.1

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	28.61	177.00	Passed
Pm+Pl+Q (TOTAL)	164.45	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.





Input, Nozzle Desc: N72"

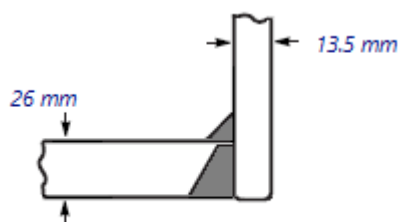
From: 80

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13000.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N72"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tt	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N7_2".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N7_2"

Intermediate Calc. for nozzle/shell welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

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Input, Nozzle Desc: N52"

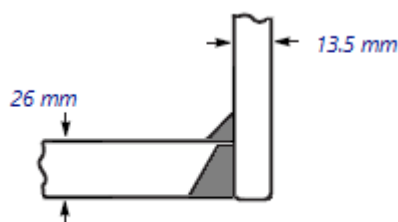
From: 90

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Cylindrical Shell	D	5500.00	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15050.00	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N52"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 2753 / (138 \cdot 1 - 0.6 \cdot 1.15)$
 $= 23.0000 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	123.7240	mm
Parallel to Vessel Wall	Rn+tn+tt	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N52".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 26.0000	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 26.0000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(26, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N52"

Intermediate Calc. for nozzle/shell welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.2746 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.2746 mm

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Input, Nozzle Desc: N210"

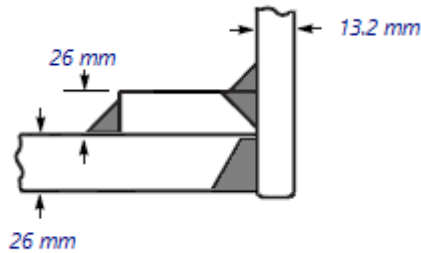
From: 100

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	5500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	+N
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	460.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		93.4750	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Reinforcement CALCULATION, Description: N210"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	9.710 in.
Actual Thickness Used in Calculation	0.520 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 126 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 1.2353 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	505.2934 mm
Parallel to Vessel Wall, opening length	d	252.6467 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		51.5041 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

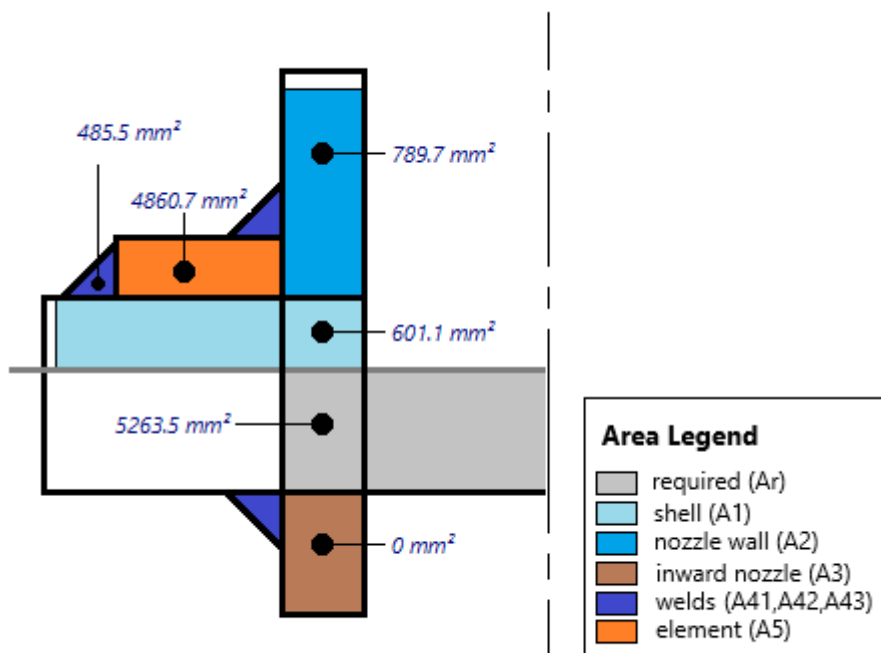


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5263.513	NA	NA
Area in Shell	A1	601.139	NA	NA
Area in Nozzle Wall	A2	789.736	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	485.506	NA	NA
Area in Element	A5	4860.700	NA	NA
TOTAL AREA AVAILABLE	Atot	6737.081	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	403.3243	26.0000 mm
Based on given Pad Diameter:	460.0000	18.1178 mm
Based on the Estimated Diameter Limit:	503.7059	14.6848 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (253 \cdot 20.6 \cdot 1 + 2 \cdot 10.2 \cdot 20.6 \cdot 1 \cdot (1 - 0.86))$
 $= 5263.513 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 253(1 \cdot 23 - 1 \cdot 20.6) - 2 \cdot 10.2$
 $(1 \cdot 23 - 1 \cdot 20.6) \cdot (1 - 0.86)$
 $= 601.139 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 51.5) \cdot (10.2 - 1.24) \cdot 0.86$

= 789.736 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - c_{an}/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1$
 = 485.506 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(460 - 273) \cdot 26 \cdot 1$
 = 4860.700 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2353 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.8711 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.8711 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 11.1026 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[11.1, max(25.9, 4.5)]
 = 11.1026 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(4.24, 11.1)
 = 11.1026 mm

Available Nozzle Neck Thickness = 13.2016 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.5,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.5 MPa	Passed
Occasional	: 6.8,	Allowable	: 156.9 MPa	Passed
Shear	: 16.7,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 26, tr = 20.6, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-34 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.2, tr = 1.24, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.12$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-20 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -40 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -40 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15/1.96 = 0.585$

Weld Size Calculations, Description: N210"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.2016 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$14.1400 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5264 - 601 + 2 \cdot 10.2 \cdot 0.86 \cdot (1 \cdot 23 - 20.6)) \cdot 138)$
 $= 649148.69 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (790 + 4861 + 486 - 0 \cdot 0.86) \cdot 138$
 $= 846687.94 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (790 + 0 + 85.5 + (401)) \cdot 138$
 $= 176141.89 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (790 + 0 + 486 + 4861 + (401)) \cdot 138$
 $= 902056.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 273 \cdot 10 \cdot 0.49 \cdot 118$
 $= 247968. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 460 \cdot 20 \cdot 0.49 \cdot 138$
 $= 977116. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 131) \cdot (13.2 - 3) \cdot 0.7 \cdot 118$
 $= 347881. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 273 \cdot 26 \cdot 0.74 \cdot 138$
 $= 1138700. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 273 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 1007311. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (977116 + 347881) = 1324996 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (247968 + 1138699 + 1007311 + 0) = 2393979 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (977116 + 1007311 + 0) = 1984426 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1324996 N , must exceed W = 649148 N or W1 = 846687 N
 Path 2-2 = 2393979 N , must exceed W = 649148 N or W2 = 176141 N
 Path 3-3 = 1984426 N , must exceed W = 649148 N or W3 = 902056 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.8751 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 227.8750 mm

Input Echo, WRC107/537 Item 1, Description: N210" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	9899.999	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow
Diameter Basis for Nozzle	Nbasis	ID



Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	246.647	mm
Nozzle Thickness	Tn	13.202	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	26.000	mm
Diameter of Reinforcing Pad	Dpad	460.000	mm
Design Internal Pressure	Dp	1.147	MPa
Include Pressure Thrust		No	

External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-15999.3	N
Longitudinal Shear (SUS)	(V1) V1	14141.5	N
Circumferential Shear (SUS)	(Vc) V2	14141.5	N
Circumferential Moment (SUS)	(Mc) M1	9280511.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	9280511.0	N-mm
Torsional Moment (SUS)	Mt	12004206.0	N-mm

Use Interactive Control

No

WRC107 Version

Version March 1979

Include Pressure Stress Indices per Div. 2

No

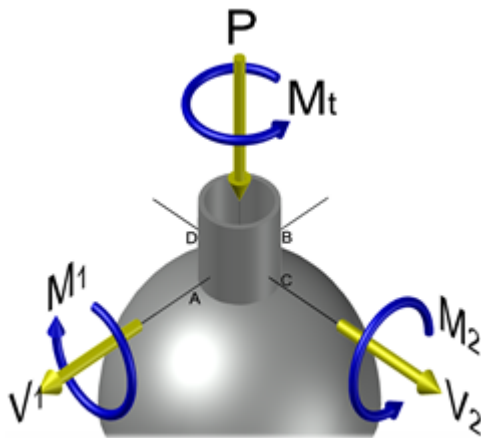
Compute Pressure Stress per WRC-368

No

Local Loads applied at end of Nozzle/Attachment

No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 273.05 + 2 \cdot 1.65 \cdot \sqrt{(4964.5 (26.0 - 3.0))} \\
 &= 1388.155 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUS tained loads:

Radial Load	P	-15999.3	N
Circumferential Shear	(VC) V2	14141.5	N
Longitudinal Shear	(VL) V1	14141.5	N
Circumferential Moment	(MC) M1	9280511.0	N-mm
Longitudinal Moment	(ML) M2	9280511.0	N-mm
Torsional Moment	MT	12004206.0	N-mm

Dimensionless Param: U = 0.28 TAU = 12.88 RHO = 4.00 (4.80)

**Dimensionless Loads for Spherical Shells at Attachment Junction:**

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.05792	(A,B,C,D)
$M(x) / P$	SP 4	0.03079	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.18137	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.10294	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.18137	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.10294	(A,B,C,D)
$N(y) * T / P$	SP 4	0.35652	(A,B,C,D)
$M(y) / P$	SP 4	0.14856	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.25255	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	0.59902	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.25255	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	0.59902	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$ **Stresses in the Vessel at the Attachment Junction (MPa)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4
Rad.	Bend. P	1.2	-1.2	1.2	-1.2	1.2	-1.2	1.2	-1.2
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.4	-1.4	1.4	1.4
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-4.8	4.8	4.8	-4.8
Rad.	Memb. ML	-1.4	-1.4	1.4	1.4	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-4.8	4.8	4.8	-4.8	0.0	0.0	0.0	0.0
Tot. Rad.	Str.	-4.6	2.6	7.9	-4.3	-4.6	2.6	7.9	-4.3
Tang.	Memb. P	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Tang.	Bend. P	5.9	-5.9	5.9	-5.9	5.9	-5.9	5.9	-5.9
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-2.0	-2.0	2.0	2.0
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-28.1	28.1	28.1	-28.1
Tang.	Memb. ML	-2.0	-2.0	2.0	2.0	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-28.1	28.1	28.1	-28.1	0.0	0.0	0.0	0.0
Tot. Tang.	Str.	-21.8	22.6	38.4	-29.7	-21.8	22.6	38.4	-29.7
	Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
	Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
	Shear MT	2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Tot. Shear		2.8	2.8	1.4	1.4	1.4	1.4	2.8	2.8
Str. Int.		22.2	23.0	38.5	29.8	21.9	22.7	38.7	30.0

Unitless Prm: $U = 0.68$ TAU = 0.00 (22.05) RHO = 0.00 (2.25)**Dimensionless Loads for Spherical Shells at Pad edge:**

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SR 2	0.10223	(A,B,C,D)
$M(x) / P$	SR 2	0.05760	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.11699	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SR 3	0.15158	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.11699	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SR 3	0.15158	(A,B,C,D)
$N(y) * T / P$	SR 2	0.03059	(A,B,C,D)
$M(y) / P$	SR 2	0.01779	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SR 3	0.03533	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SR 3	0.04479	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SR 3	0.03533	(A,B,C,D)

$M(y) * \sqrt{R_m * T} / ML$ SR 3 0.04479 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Rad. Bend. P		10.5	-10.5	10.5	-10.5	10.5	-10.5	10.5	-10.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-6.1	-6.1	6.1	6.1
Rad. Bend. MC		0.0	0.0	0.0	0.0	-47.2	47.2	47.2	-47.2
Rad. Memb. ML		-6.1	-6.1	6.1	6.1	0.0	0.0	0.0	0.0
Rad. Bend. ML		-47.2	47.2	47.2	-47.2	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-39.7	33.8	66.8	-48.5	-39.7	33.8	66.8	-48.5
Tang. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P		3.2	-3.2	3.2	-3.2	3.2	-3.2	3.2	-3.2
Tang. Memb. MC		0.0	0.0	0.0	0.0	-1.8	-1.8	1.8	1.8
Tang. Bend. MC		0.0	0.0	0.0	0.0	-13.9	13.9	13.9	-13.9
Tang. Memb. ML		-1.8	-1.8	1.8	1.8	0.0	0.0	0.0	0.0
Tang. Bend. ML		-13.9	13.9	13.9	-13.9	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-11.6	9.8	19.9	-14.4	-11.6	9.8	19.9	-14.4
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Tot. Shear		2.4	2.4	0.7	0.7	0.7	0.7	2.4	2.4
Str. Int.		39.9	34.0	66.8	48.5	39.7	33.8	66.9	48.7

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Rad. Pl (SUS)		-1.0	-1.0	1.8	1.8	-1.0	-1.0	1.8	1.8
Rad. Q (SUS)		-3.6	3.6	6.1	-6.1	-3.6	3.6	6.1	-6.1
Long. Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Long. Pl (SUS)		0.4	0.4	4.4	4.4	0.4	0.4	4.4	4.4
Long. Q (SUS)		-22.2	22.2	34.1	-34.1	-22.2	22.2	34.1	-34.1
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear Q (SUS)		2.1	2.1	2.1	2.1	2.1	2.1	2.1	2.1
Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Pm+Pl (SUS)		58.6	58.6	62.5	62.5	58.6	58.6	62.5	62.5
Pm+Pl+Q (Total)		53.8	80.9	96.5	53.8	53.5	80.7	96.6	54.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	57.98	138.00	Passed



Pm+Pl (SUS)	62.50	207.00	Passed
Pm+Pl+Q (TOTAL)	96.64	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Rad.	Pl (SUS)	-3.0	-3.0	9.2	9.2	-3.0	-3.0	9.2	9.2
Rad.	Q (SUS)	-36.7	36.7	57.7	-57.7	-36.7	36.7	57.7	-57.7
Long.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Long.	Pl (SUS)	-0.9	-0.9	2.8	2.8	-0.9	-0.9	2.8	2.8
Long.	Q (SUS)	-10.7	10.7	17.2	-17.2	-10.7	10.7	17.2	-17.2
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear	Q (SUS)	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6
Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Pm+Pl (SUS)		122.9	122.9	132.8	132.8	122.9	122.9	132.8	132.8
Pm+Pl+Q (Total)		112.1	157.5	190.4	109.1	111.9	157.3	190.5	109.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.52	138.00	Passed
Pm+Pl (SUS)	132.80	207.00	Passed
Pm+Pl+Q (TOTAL)	190.47	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N46"

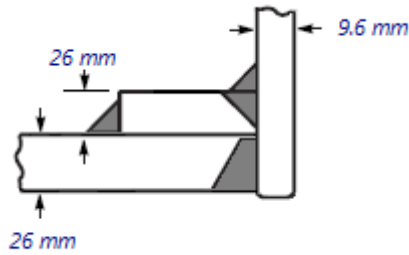
From: 100

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	5500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	+N
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	250.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		40.8625	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N46"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 77.5 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.7582 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	326.1440 mm
Parallel to Vessel Wall, opening length	d	163.0720 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5030 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, Sn/Sv)$
 $= \min(1, 118/138)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, Sp/Sv)$
 $= \min(1, 138/138)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.86, 1)$
 $= 0.855$

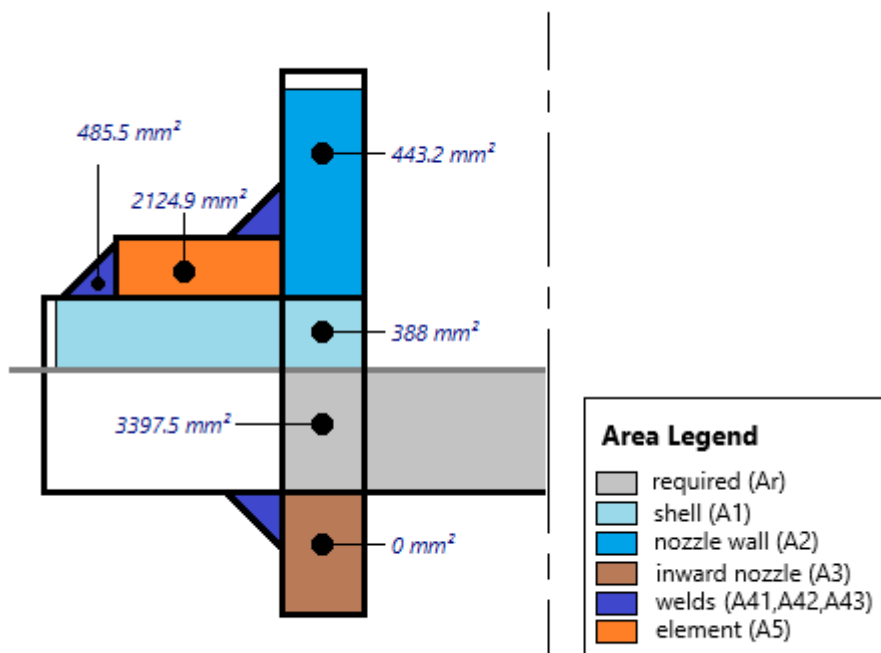


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3397.458	NA	NA
Area in Shell	A1	387.996	NA	NA
Area in Nozzle Wall	A2	443.195	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	485.506	NA	NA
Area in Element	A5	2124.850	NA	NA
TOTAL AREA AVAILABLE	Atot	3441.548	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 71.98 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	248.3043	26.0000 mm
Based on given Pad Diameter:	250.0000	25.4605 mm
Based on the Estimated Diameter Limit:	324.5565	14.0972 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (163 \cdot 20.6 \cdot 1 + 2 \cdot 6.6 \cdot 20.6 \cdot 1 \cdot (1 - 0.86))$
 $= 3397.458 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 163(1 \cdot 23 - 1 \cdot 20.6) - 2 \cdot 6.6$
 $(1 \cdot 23 - 1 \cdot 20.6) \cdot (1 - 0.86)$
 $= 387.996 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 / \sin(\alpha3)$
 $= (2 \cdot 42.5) \cdot (6.6 - 0.76) \cdot 0.86 / \sin(73.4)$

= 443.195 mm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can / 0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1$
 = 485.506 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (Nozzle OD)) \cdot (\min(tp, Tlwp, te)) \cdot fr_4$
 = $(259 - 177) \cdot 26 \cdot 1$
 = 2124.850 mm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 9.6, tr = 0.76, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 9.6, tr = 0.76, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: D

Govrn. thk, tg = 9.6, tr = 0.76, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: D

Govrn. thk, tg = 26, tr = 20.6, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve D	-34 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-40 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 9.6, tr = 0.76, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.11$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -40 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -40 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 5.10 = 0.225$

Weld Size Calculations, Description: N46"



Intermediate Calc. for nozzle/shell Welds Tmin 6.6012 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.6208 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	14.1400 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (3397 - 388 + 2 \cdot 6.6 \cdot 0.86 \cdot (1 \cdot 23 - 20.6)) \cdot 138)$
 $= 419020.72 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (443 + 2125 + 486 - 0 \cdot 0.86) \cdot 138$
 $= 421354.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (443 + 0 + 85.5 + (260)) \cdot 138$
 $= 108782.16 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (443 + 0 + 486 + 2125 + (260)) \cdot 138$
 $= 457181.84 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.14/2.0) \cdot 177 \cdot 10 \cdot 0.49 \cdot 118$
 $= 160700. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.14/2.0) \cdot 250 \cdot 20 \cdot 0.49 \cdot 138$
 $= 531041. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.14 \cdot 85) \cdot (9.6 - 3) \cdot 0.7 \cdot 118$
 $= 145600. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.14/2) \cdot 177 \cdot 26 \cdot 0.74 \cdot 138$
 $= 737957. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.14/2.0) \cdot 177 \cdot (26 - 3) \cdot 0.74 \cdot 138$
 $= 652808. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (531041 + 145600) = 676641 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$



$$= (160700 + 737957 + 652808 + 0) = 1551465 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (531041 + 652808 + 0) = 1183849 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 676641 N , must exceed W = 419020 N or W1 = 421354 N

Path 2-2 = 1551465 N , must exceed W = 419020 N or W2 = 108782 N

Path 3-3 = 1183849 N , must exceed W = 419020 N or W3 = 457181 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N46"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869 in.
Actual Thickness Used in Calculation	0.378 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]

$$= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P) \text{ per Appendix 1-4(c)}$$

$$= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$$

$$= 20.5925 \text{ mm}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (Sn \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 1.15 \cdot 77.5 / (118 \cdot 1 - 0.6 \cdot 1.15)$$

$$= 0.7582 \text{ mm}$$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	310.1452 mm
Parallel to Vessel Wall, opening length	d	155.0726 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		42.5030 mm

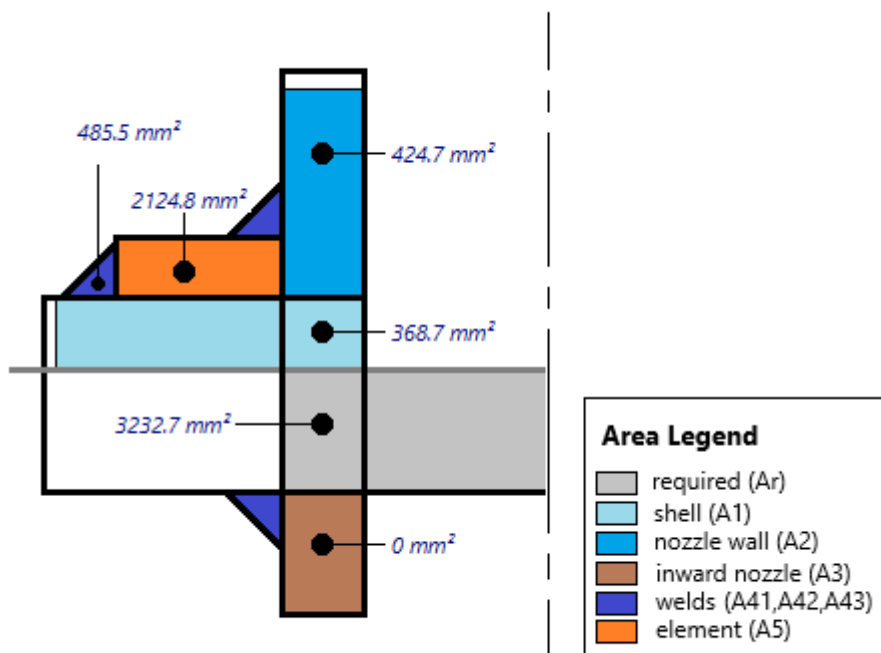


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	3232.731	NA	NA
Area in Shell	A1	368.738	NA	NA
Area in Nozzle Wall	A2	424.696	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	485.506	NA	NA
Area in Element	A5	2124.850	NA	NA
TOTAL AREA AVAILABLE	Atot	3403.789	NA	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	243.4208	26.0000 mm
Based on given Pad Diameter:	250.0000	23.9069 mm
Based on the Estimated Diameter Limit:	308.5577	13.9275 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155 \cdot 20.6 \cdot 1 + 2 \cdot 6.6 \cdot 20.6 \cdot 1 \cdot (1 - 0.86))$
 $= 3232.731 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 155(1 \cdot 23 - 1 \cdot 20.6) - 2 \cdot 6.6$
 $(1 \cdot 23 - 1 \cdot 20.6) \cdot (1 - 0.86)$
 $= 368.738 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 42.5) \cdot (6.6 - 0.76) \cdot 0.86$

= 424.696 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
 = $10^2 \cdot 0.86 + (0)^2 \cdot 0.86 + 20^2 \cdot 1$
 = 485.506 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (Nozzle\ OD)) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
 = $(250 - 168) \cdot 26 \cdot 1$
 = 2124.850 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.7582 mm
Wall Thickness per UG16(b),	tr16b = 4.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.8711 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.8711 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm
Wall Thickness per table UG-45	tb3 = 9.2200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[9.22, max(25.9, 4.5)]
 = 9.2200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.76, 9.22)
 = 9.2200 mm

Available Nozzle Neck Thickness = 9.6012 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 39.8,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 255.2 MPa	Passed
Occasional	: 6.5,	Allowable	: 156.9 MPa	Passed
Shear	: 24.8,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N46"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.6012 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.6208 = $0.7 \cdot t_{min}$	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	14.1400 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 = max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
 = max(0, (3233 - 369 + 2 • 6.6 • 0.86 • (1 • 23 - 20.6))138)
 = 398947.75 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 = (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
 = (425 + 2125 + 486 - 0 • 0.86) • 138
 = 418801.53 N

Weld Load [W2]:
 = (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv



$$= (425 + 0 + 85.5 + (260)) \cdot 138$$

$$= 106229.51 \text{ N}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$$

$$= (425 + 0 + 486 + 2125 + (260)) \cdot 138$$

$$= 454629.12 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.14/2.0) \cdot 168 \cdot 10 \cdot 0.49 \cdot 118$$

$$= 152817. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.14/2.0) \cdot 250 \cdot 20 \cdot 0.49 \cdot 138$$

$$= 531041. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.14 \cdot 80.8) \cdot (9.6 - 3) \cdot 0.7 \cdot 118$$

$$= 138458. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.14/2) \cdot 168 \cdot 26 \cdot 0.74 \cdot 138$$

$$= 701757. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.14/2.0) \cdot 168 \cdot (26 - 3) \cdot 0.74 \cdot 138$$

$$= 620785. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (531041 + 138458) = 669499 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (152817 + 701757 + 620785 + 0) = 1475359 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (531041 + 620785 + 0) = 1151825 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 669498 N	, must exceed W = 398947 N	or W1 = 418801 N
Path 2-2 = 1475359 N	, must exceed W = 398947 N	or W2 = 106229 N
Path 3-3 = 1151825 N	, must exceed W = 398947 N	or W3 = 454629 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1.15 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 27.4572 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 354.7215 mm

Warning:

The equations for nozzle local stress analysis pertain to radial nozzles only and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N46" :



Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	9899.999	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	149.073	mm
Nozzle Thickness	Tn	9.601	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Thickness of Reinforcing Pad	Tpad	26.000	mm
Diameter of Reinforcing Pad	Dpad	250.000	mm
Design Internal Pressure	Dp	1.147	MPa
Include Pressure Thrust		No	

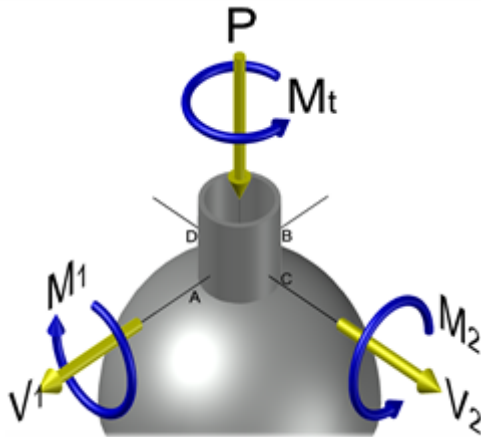
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-9599.6	N
Longitudinal Shear (SUS)	(V1) V1	8484.9	N
Circumferential Shear (SUS)	(Vc) V2	8484.9	N
Circumferential Moment (SUS)	(Mc) M1	3340974.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	3340974.0	N-mm
Torsional Moment (SUS)	Mt	4321525.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(4964.5 (26.0 - 3.0))}$$

$$= 1283.380 \text{ mm}$$

WRC 107 Stress Calculation for SUSTAINED loads:

Radial Load		P	-9599.6	N
Circumferential Shear	(VC)	V2	8484.9	N
Longitudinal Shear	(VL)	V1	8484.9	N
Circumferential Moment	(MC)	M1	3340974.0	N-mm
Longitudinal Moment	(ML)	M2	3340974.0	N-mm
Torsional Moment		MT	4321525.0	N-mm

Dimensionless Param: U = 0.17 TAU = 12.25 RHO = 4.00 (7.42)

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 4	0.07600	(A,B,C,D)
$M(x) / P$	SP 4	0.04211	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.28360	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 4	0.17247	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.28360	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 4	0.17247	(A,B,C,D)
$N(y) * T / P$	SP 4	0.35584	(A,B,C,D)
$M(y) / P$	SP 4	0.22312	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 4	0.10471	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 4	1.06888	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 4	0.10471	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 4	1.06888	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad. Bend.	P	1.0	-1.0	1.0	-1.0	1.0	-1.0	1.0	-1.0
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-0.8	-0.8	0.8	0.8
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-2.9	2.9	2.9	-2.9
Rad. Memb.	ML	-0.8	-0.8	0.8	0.8	0.0	0.0	0.0	0.0
Rad. Bend.	ML	-2.9	2.9	2.9	-2.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-2.4	1.4	5.0	-2.8	-2.4	1.4	5.0	-2.8



Tang. Memb. P	1.4	1.4	1.4	1.4	1.4	1.4	1.4	1.4
Tang. Bend. P	5.4	-5.4	5.4	-5.4	5.4	-5.4	5.4	-5.4
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.3	-0.3	0.3	0.3
Tang. Bend. MC	0.0	0.0	0.0	0.0	-18.1	18.1	18.1	-18.1
Tang. Memb. ML	-0.3	-0.3	0.3	0.3	0.0	0.0	0.0	0.0
Tang. Bend. ML	-18.1	18.1	18.1	-18.1	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-11.6	13.8	25.1	-21.7	-11.6	13.8	25.1	-21.7
Shear VC	0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Tot. Shear	2.6	2.6	1.3	1.3	1.3	1.3	2.6	2.6
Str. Int.	12.3	14.4	25.2	21.8	11.8	14.0	25.5	22.1

Unitless Prm: U = 0.37 TAU = 0.00 (18.44) RHO = 0.00 (3.48)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.16850	(A,B,C,D)
M(x) / P	SR 2	0.11510	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.15943	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.36861	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.15943	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.36861	(A,B,C,D)
N(y) * T / P	SR 2	0.05056	(A,B,C,D)
M(y) / P	SR 2	0.03448	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.04911	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SR 3	0.11013	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SR 3	0.04911	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SR 3	0.11013	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.1	3.1	3.1	3.1	3.1	3.1	3.1	3.1
Rad. Bend. P		12.5	-12.5	12.5	-12.5	12.5	-12.5	12.5	-12.5
Rad. Memb. MC		0.0	0.0	0.0	0.0	-3.0	-3.0	3.0	3.0
Rad. Bend. MC		0.0	0.0	0.0	0.0	-41.3	41.3	41.3	-41.3
Rad. Memb. ML		-3.0	-3.0	3.0	3.0	0.0	0.0	0.0	0.0
Rad. Bend. ML		-41.3	41.3	41.3	-41.3	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-28.7	28.9	59.9	-47.8	-28.7	28.9	59.9	-47.8
Tang. Memb. P		0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Tang. Bend. P		3.8	-3.8	3.8	-3.8	3.8	-3.8	3.8	-3.8
Tang. Memb. MC		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Tang. Bend. MC		0.0	0.0	0.0	0.0	-12.3	12.3	12.3	-12.3
Tang. Memb. ML		-0.9	-0.9	0.9	0.9	0.0	0.0	0.0	0.0
Tang. Bend. ML		-12.3	12.3	12.3	-12.3	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-8.6	8.6	17.9	-14.3	-8.6	8.6	17.9	-14.3
Shear VC		0.9	0.9	-0.9	-0.9	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.9	-0.9	0.9	0.9
Shear MT		1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Tot. Shear		2.9	2.9	1.0	1.0	1.0	1.0	2.9	2.9



Str. Int.	29.1	29.3	59.9	47.8	28.8	28.9	60.1	48.1
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WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Rad.	Pl (SUS)	-0.5	-0.5	1.1	1.1	-0.5	-0.5	1.1	1.1
Rad.	Q (SUS)	-1.9	1.9	3.9	-3.9	-1.9	1.9	3.9	-3.9
Long.	Pm (SUS)	58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Long.	Pl (SUS)	1.1	1.1	1.7	1.7	1.1	1.1	1.7	1.7
Long.	Q (SUS)	-12.7	12.7	23.4	-23.4	-12.7	12.7	23.4	-23.4
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	2.0	2.0	2.0	2.0	2.0	2.0	2.0	2.0
Pm (SUS)		58.0	58.0	58.0	58.0	58.0	58.0	58.0	58.0
Pm+Pl (SUS)		59.3	59.3	60.1	60.1	59.3	59.3	60.1	60.1
Pm+Pl+Q (Total)		56.3	72.4	83.2	55.3	55.8	72.0	83.5	55.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	57.98	138.00	Passed
Pm+Pl (SUS)	60.12	207.00	Passed
Pm+Pl+Q (TOTAL)	83.46	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Rad.	Pl (SUS)	0.1	0.1	6.0	6.0	0.1	0.1	6.0	6.0
Rad.	Q (SUS)	-28.8	28.8	53.9	-53.9	-28.8	28.8	53.9	-53.9
Long.	Pm (SUS)	123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Long.	Pl (SUS)	-0.0	-0.0	1.8	1.8	-0.0	-0.0	1.8	1.8
Long.	Q (SUS)	-8.6	8.6	16.1	-16.1	-8.6	8.6	16.1	-16.1
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.9	0.9	-0.9	-0.9	-0.9	-0.9	0.9	0.9
Shear	Q (SUS)	1.9	1.9	1.9	1.9	1.9	1.9	1.9	1.9
Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Pm+Pl (SUS)		124.5	124.5	129.8	129.8	124.5	124.5	129.8	129.8
Pm+Pl+Q (Total)		115.3	152.8	183.4	109.3	115.0	152.4	183.6	109.5

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.52	138.00	Passed
Pm+Pl (SUS)	129.76	207.00	Passed
Pm+Pl+Q (TOTAL)	183.61	414.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N62"

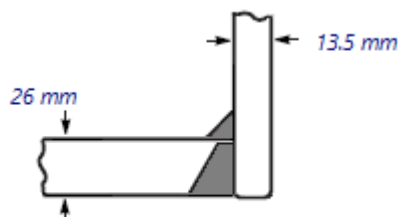
From: 100

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	5500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.6670	mm
Parallel to Vessel Wall	Rn+tn+t	63.3335	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-41 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 1.96 = 0.585$

Weld Size Calculations, Description: N62"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N62"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	123.7240	mm
Parallel to Vessel Wall	Rn+tn+t	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N62".
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 25.8711	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 25.8711	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(25.9, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Weld Size Calculations, Description: N62"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 12.4760 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.7402 mm

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Input, Nozzle Desc: N82"

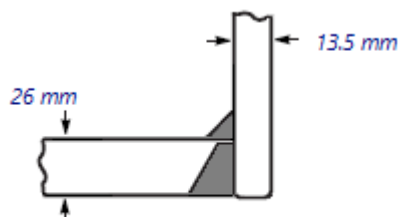
From: 100

Pressure for Reinforcement Calculations	P	1.1472	MPa
Temperature for Internal Pressure	Temp	85	°C
Parent Material		SA-516 70	+N
Parent Allowable Stress at Temperature	Sv	138.00	MPa
Parent Allowable Stress At Ambient	Sva	138.00	MPa
Inside Diameter of Elliptical Head	D	5500.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	26.0000	mm
Head Internal Corrosion Allowance	cas	3.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	1500.0000	mm
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		45.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	13.4620	mm
Flange Material		SA-105	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (to the nozzle).



Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Inside Diameter Used in Calculation 2.000 in.
 Actual Thickness Used in Calculation 0.530 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot Sv \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (Sn \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777$ mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	126.6670	mm
Parallel to Vessel Wall	Rn+tn+t	63.3335	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N82".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:B, Curve:D)

Govrn. thk, tg = 13.5, tr = 0.28, c = 3 mm, E* = 1
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.027$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1.15 / 10.20 = 0.112$

Weld Size Calculations, Description: N82"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
 exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N82"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.530	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K1 / (2 \cdot S \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 1.15 \cdot 5506 \cdot 0.9 / (2 \cdot 138 \cdot 1 - 0.2 \cdot 1.15)$
 $= 20.5925 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.15 \cdot 28.4 / (118 \cdot 1 - 0.6 \cdot 1.15)$
 $= 0.2777 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	123.7240	mm
Parallel to Vessel Wall	Rn+tn+t	61.8620	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	26.1550	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N62".
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.2777	mm
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 25.8711	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 25.8711	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm
Wall Thickness per table UG-45	tb3	= 7.8000	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[7.8, \max(25.9, 4.5)]$
 $= 7.8000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(3.28, 7.8)$
 $= 7.8000 \text{ mm}$

Available Nozzle Neck Thickness = 13.4620 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	15.8,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	279.2 MPa	Passed
Occasional	:	1.3,	Allowable	:	156.9 MPa	Passed
Shear	:	13.3,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Weld Size Calculations, Description: N82"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.4620	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 12.4760 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 239.7402 mm

Warning:
The equations for nozzle local stress analysis pertain to radial nozzles only



and this nozzle is not specified as radial. Please check the specific analysis code for all assumptions and limitations.

Input Echo, WRC107/537 Item 1, Description: N82" :

Diameter Basis for Vessel	Vbasis	ID	
Cylindrical or Spherical Vessel	Cylsph	Spherical	
Internal Corrosion Allowance	Cas	3.0000	mm
Vessel Diameter	Dv	9899.999	mm
Vessel Thickness	Tv	26.000	mm
Design Temperature	T1	85.0	C
Vessel Material		SA-516 70	
Vessel UNS Number		K02700	
Vessel Cold S.I. Allowable	Smc	138.00	MPa
Vessel Hot S.I. Allowable	Smh	138.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round	
WRC107 Attachment Classification	Holsol	Hollow	
Diameter Basis for Nozzle	Nbasis	ID	
Corrosion Allowance for Nozzle	Can	3.0000	mm
Nozzle Diameter	Dn	50.800	mm
Nozzle Thickness	Tn	13.462	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa
Design Internal Pressure	Dp	1.147	MPa
Include Pressure Thrust		No	

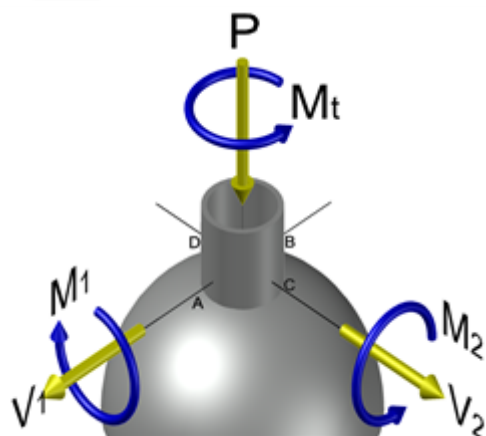
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-3199.9	N
Longitudinal Shear (SUS)	(V1) V1	2828.3	N
Circumferential Shear (SUS)	(Vc) V2	2828.3	N
Circumferential Moment (SUS)	(Mc) M1	371220.7	N-mm
Longitudinal Moment (SUS)	(Ml) M2	371220.7	N-mm
Torsional Moment (SUS)	Mt	480169.5	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



Error/Warning Messages for WRC 107/537 Analysis: 1

Warning - The ratio R_o/R_m , 0.0078, must be > 0.0571 and < 0.571 .

WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 77.724 + 2 \cdot 1.65 \cdot \sqrt{4964.5 (26.0 - 3.0)}$$

$$= 1192.829 \text{ mm}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-3199.9	N
Circumferential Shear	(VC) V2	2828.3	N
Longitudinal Shear	(VL) V1	2828.3	N
Circumferential Moment	(MC) M1	371220.7	N-mm
Longitudinal Moment	(ML) M2	371220.7	N-mm
Torsional Moment	MT	480169.5	N-mm

Dimensionless Param: $U = 0.12 \text{ TAU} = 5.00 (3.21) \text{ RHO} = 2.20$

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
$N(x) * T / P$	SP 3	0.17592	(A,B,C,D)
$M(x) / P$	SP 3	0.14104	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.57348	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / MC$	SM 3	0.52922	(A,B,C,D)
$N(x) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.57348	(A,B,C,D)
$M(x) * \text{SQRT}(R_m * T) / ML$	SM 3	0.52922	(A,B,C,D)
$N(y) * T / P$	SP 3	0.21099	(A,B,C,D)
$M(y) / P$	SP 3	0.16500	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / MC$	SM 3	0.08943	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / MC$	SM 3	1.09085	(A,B,C,D)
$N(y) * T * \text{SQRT}(R_m * T) / ML$	SM 3	0.08943	(A,B,C,D)
$M(y) * \text{SQRT}(R_m * T) / ML$	SM 3	1.09085	(A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb.	P	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
Rad. Bend.	P	5.1	-5.1	5.1	-5.1	5.1	-5.1	5.1	-5.1
Rad. Memb.	MC	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Rad. Bend.	MC	0.0	0.0	0.0	0.0	-6.6	6.6	6.6	-6.6



Rad. Memb. ML	-1.2	-1.2	1.2	1.2	0.0	0.0	0.0	0.0
Rad. Bend. ML	-6.6	6.6	6.6	-6.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.	-1.6	1.3	14.0	-9.5	-1.6	1.3	14.0	-9.5
Tang. Memb. P	1.3	1.3	1.3	1.3	1.3	1.3	1.3	1.3
Tang. Bend. P	6.0	-6.0	6.0	-6.0	6.0	-6.0	6.0	-6.0
Tang. Memb. MC	0.0	0.0	0.0	0.0	-0.2	-0.2	0.2	0.2
Tang. Bend. MC	0.0	0.0	0.0	0.0	-13.6	13.6	13.6	-13.6
Tang. Memb. ML	-0.2	-0.2	0.2	0.2	0.0	0.0	0.0	0.0
Tang. Bend. ML	-13.6	13.6	13.6	-13.6	0.0	0.0	0.0	0.0
Tot. Tang. Str.	-6.5	8.7	21.0	-18.1	-6.5	8.7	21.0	-18.1
Shear VC	1.0	1.0	-1.0	-1.0	0.0	0.0	0.0	0.0
Shear VL	0.0	0.0	0.0	0.0	-1.0	-1.0	1.0	1.0
Shear MT	2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Tot. Shear	3.2	3.2	1.2	1.2	1.2	1.2	3.2	3.2
Str. Int.	8.1	9.9	21.2	18.3	6.8	8.9	22.3	19.2

WRC 107/537 Stress Summations:**Vessel Stress Summation at Attachment Junction (MPa)**

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Rad. Pl (SUS)		-0.1	-0.1	2.3	2.3	-0.1	-0.1	2.3	2.3
Rad. Q (SUS)		-1.5	1.5	11.7	-11.7	-1.5	1.5	11.7	-11.7
Long. Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Long. Pl (SUS)		1.1	1.1	1.5	1.5	1.1	1.1	1.5	1.5
Long. Q (SUS)		-7.6	7.6	19.6	-19.6	-7.6	7.6	19.6	-19.6
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		1.0	1.0	-1.0	-1.0	-1.0	-1.0	1.0	1.0
Shear Q (SUS)		2.2	2.2	2.2	2.2	2.2	2.2	2.2	2.2
Pm (SUS)		123.5	123.5	123.5	123.5	123.5	123.5	123.5	123.5
Pm+Pl (SUS)		125.2	125.2	126.5	126.5	125.2	125.2	126.5	126.5
Pm+Pl+Q (Total)		123.5	133.4	144.8	114.2	122.2	132.4	145.8	115.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	123.52	138.00	Passed
Pm+Pl (SUS)	126.46	207.00	Passed
Pm+Pl+Q (TOTAL)	145.80	414.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: